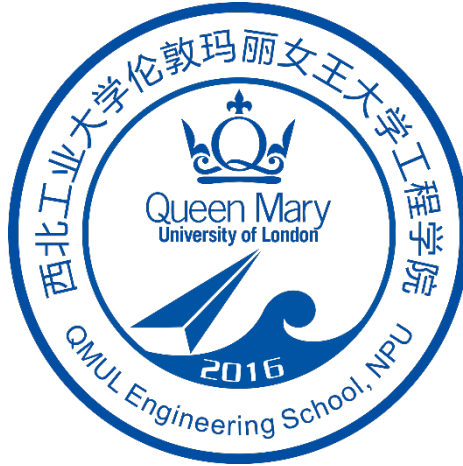


Group Report

[NXC5015: Structural Characterization]



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Group Leader: Zhang Yuchen

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Kong Xianglu	2023304256	A5
Han Yujia	2023304239	A6

* Members are ranked in no particular order

The rest of the work was completed together by the group.

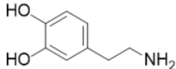
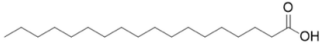
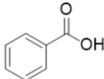
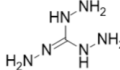
Abstract

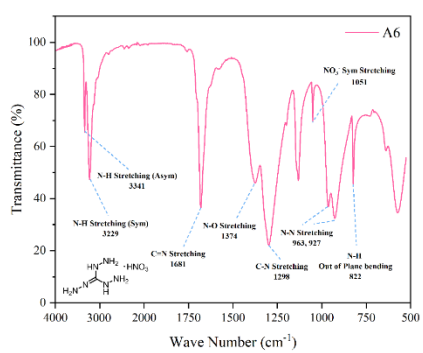
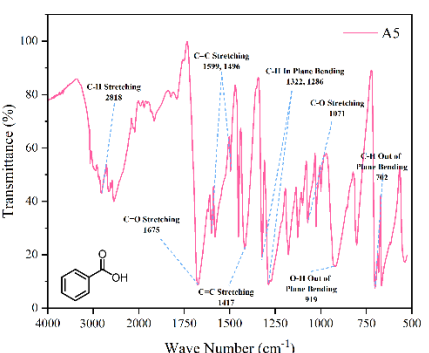
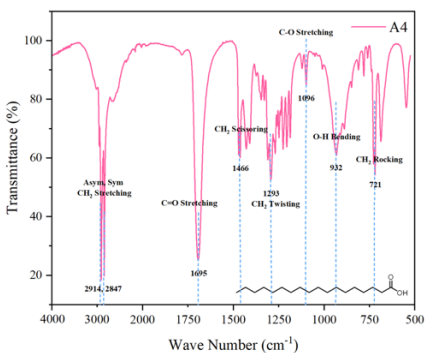
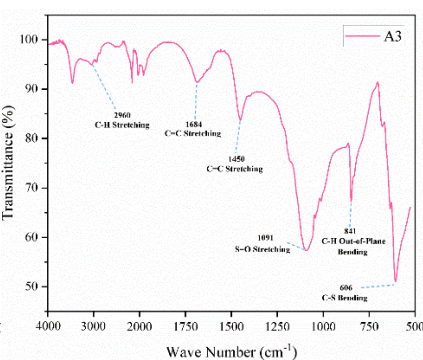
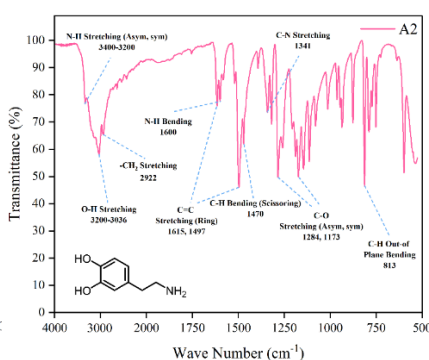
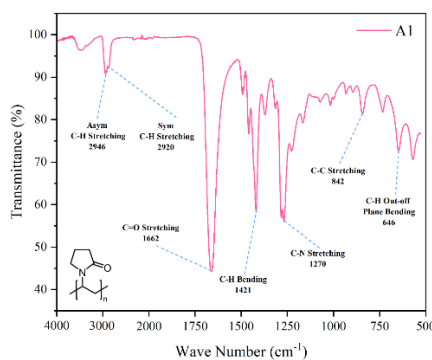
This experiment aims to learn the basic principles and experimental operation methods of infrared spectroscopy and thermal analysis (TGA/DSC), and to identify six samples through IR and thermal analysis to determine the type and structure of the samples. Infrared spectroscopy is often used to detect the functional groups contained in the samples because when the frequency of infrared light is equal to the vibration frequency of the molecule, the molecule will absorb energy and undergo a vibration transition, generating an infrared absorption peak. Different molecules have different vibration frequencies, so infrared spectroscopy is often used to detect functional groups. Thermal analysis measures and quantitatively analyzes the changes in the physical properties of the sample with temperature changes to assist in verifying the analysis results of infrared spectroscopy. During the analysis process, the reasons for the differences between the experimental spectra and the standard spectra and the reasons for the frequency differences of different vibration modes in the infrared spectrum were mainly discussed. This experiment utilized the combined effect of two detection methods, improved the accuracy of sample structure characterization, and provided an experimental method of reference for the qualitative analysis of similar substances.

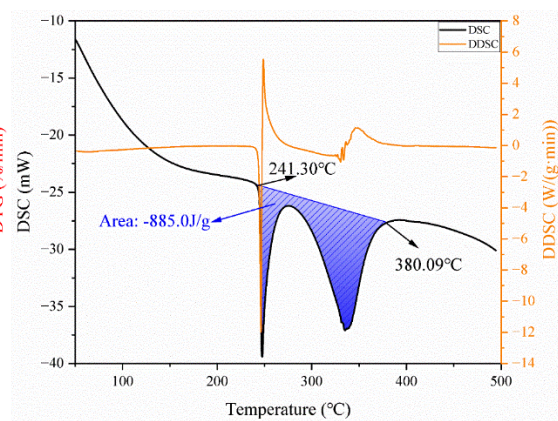
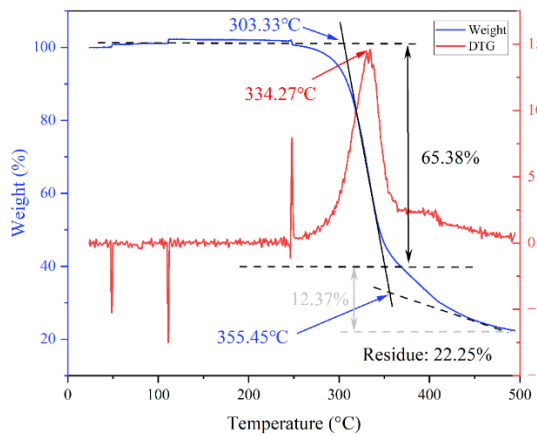
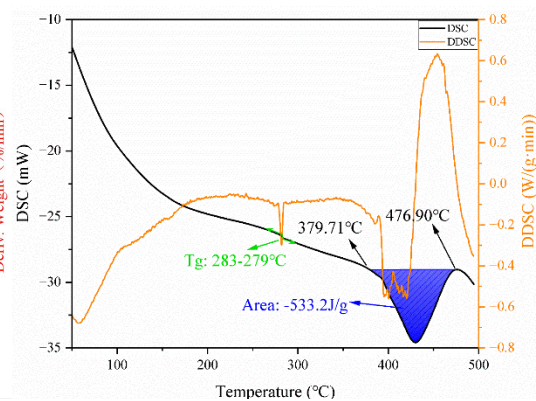
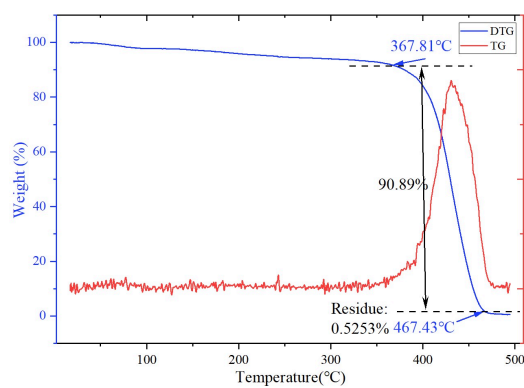
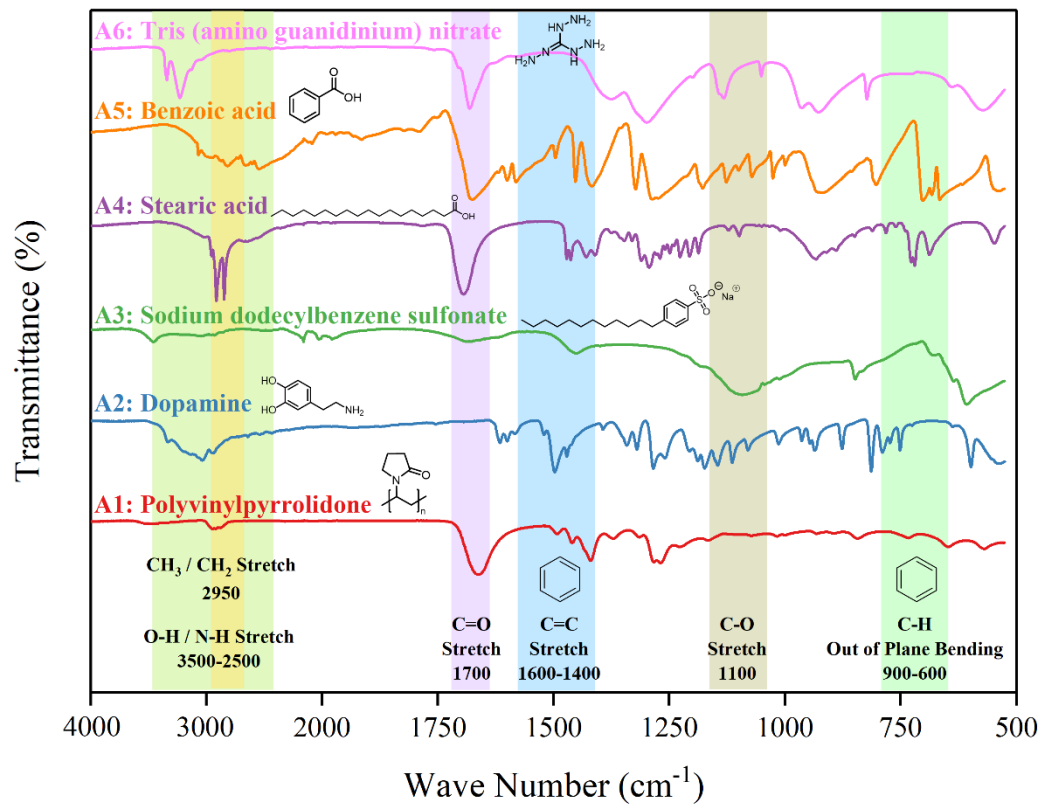
Key Words: Infrared Radiation; Thermal Analysis; FTIR; TGA; DSC.

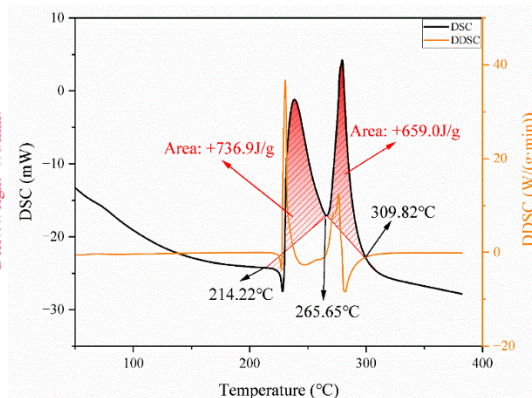
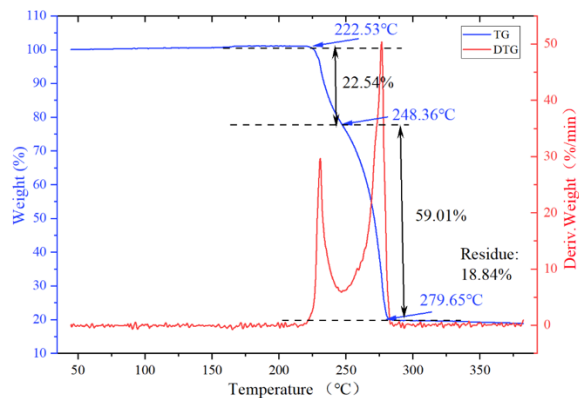
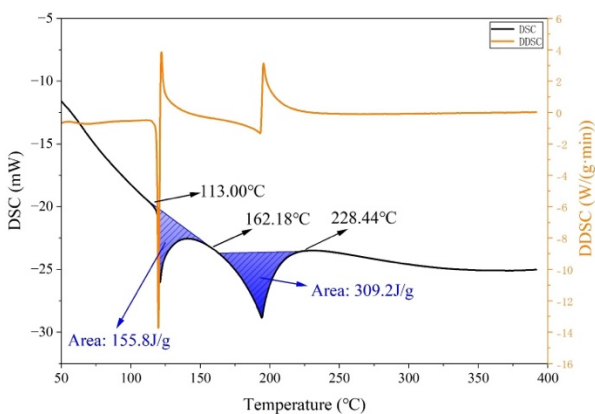
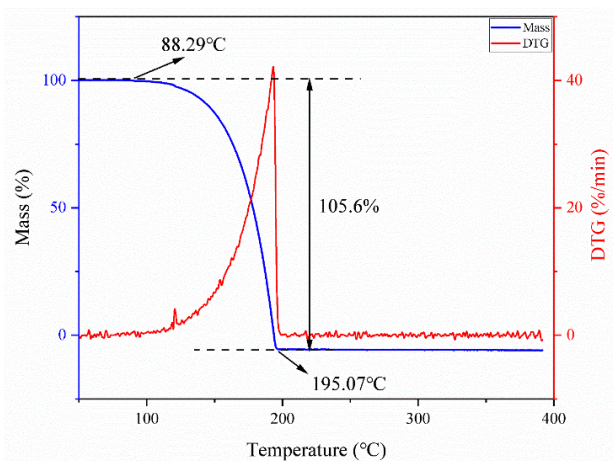
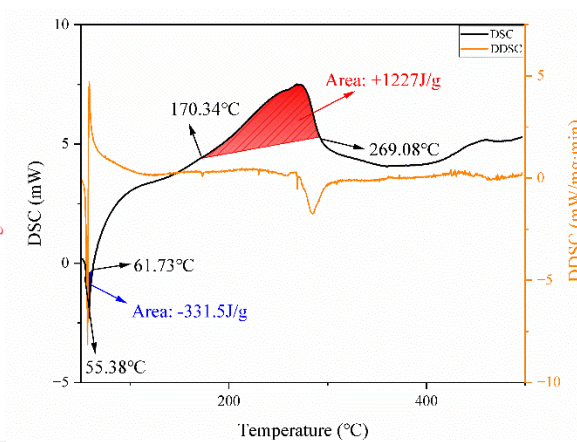
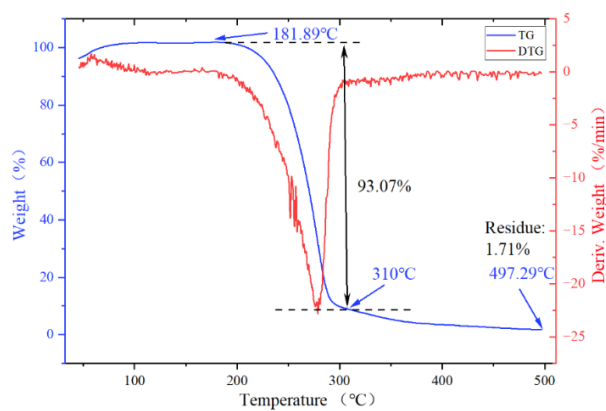
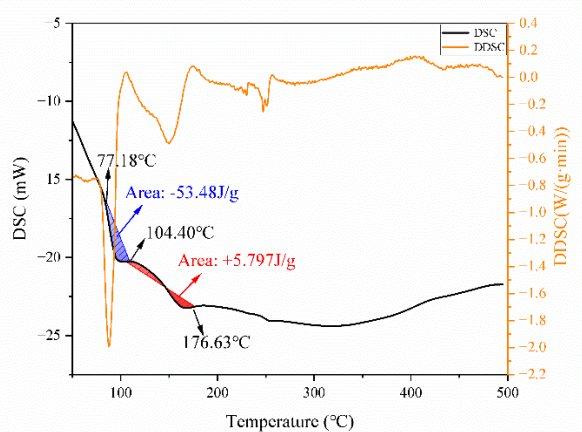
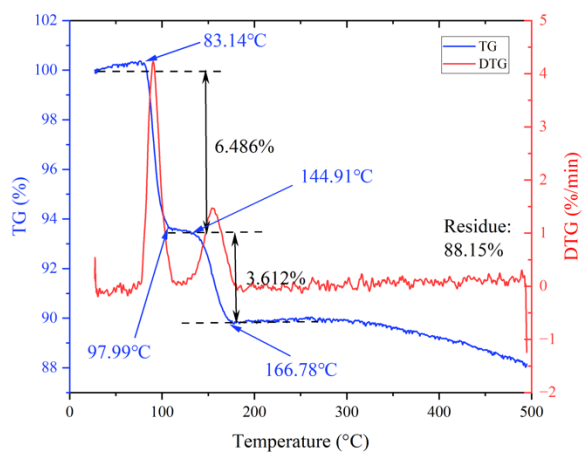
Results

Table1. Preview of experimental results

Sample	Material	Structure
A1	Polyvinylpyrrolidone	
A2	Dopamine	
A3	Sodium dodecylbenzene sulfonate	
A4	Stearic acid	
A5	Benzoic acid	
A6	Tris (amino guanidinium) nitrate	







1. Introduction

1.1 Infrared Radiation

Infrared radiation (IR) is a non-visible radiation in the electromagnetic spectrum, situated between visible light and microwaves. Its wavelength range is typically defined as 0.78 μm to 1000 μm , corresponding to frequencies of approximately 300 GHz to 385 THz. According to the classification standard of ISO 20473, infrared radiation can be further subdivided into near-infrared (NIR, 0.78–3 μm), mid-infrared (MIR, 3–50 μm), and far-infrared (FIR, 50–1000 μm). Since the energy of infrared radiation precisely matches the vibrational energy levels of chemical bonds within molecules, the selective absorption of infrared light at specific wavelengths by substances forms the physical basis of infrared spectroscopy. This characteristic makes infrared technology one of the core methods in scientific research for revealing the chemical structure of materials.

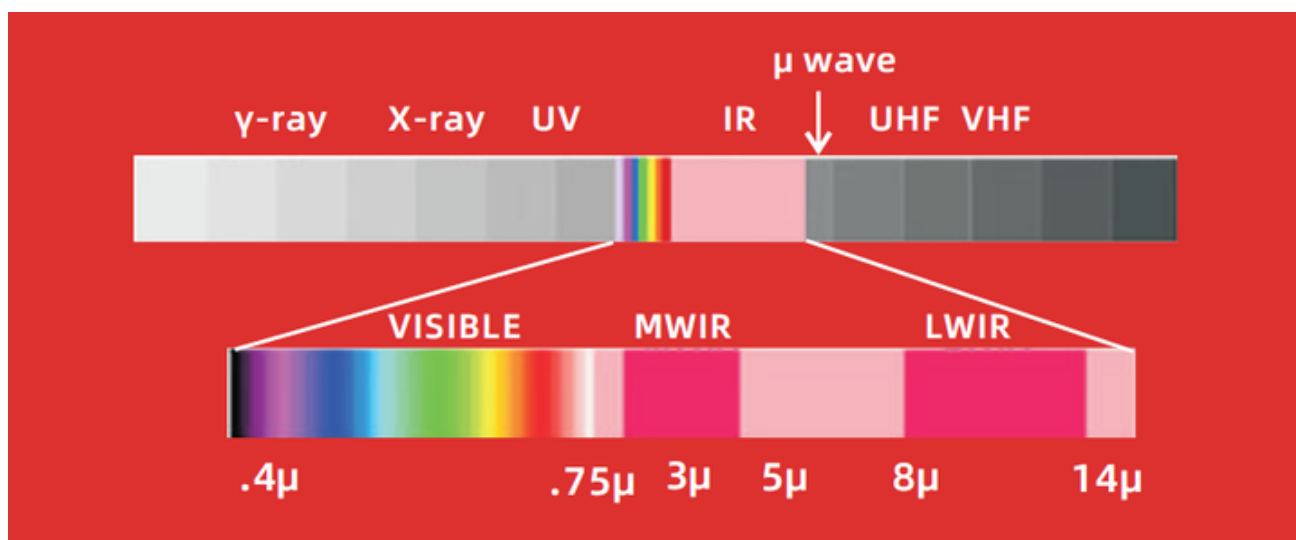


Figure 1. Position of infrared in the spectrum

Fourier Transform Infrared Spectroscopy (FTIR) is a third-generation infrared spectroscopy technique based on the principle of molecular vibrational energy level transitions. When a sample is irradiated with continuous infrared light of specific wavelengths, molecules absorb photons at specific frequencies, transitioning from lower to higher energy levels. By detecting the sample's absorption of infrared light at specific wavelengths, its "molecular fingerprint" spectrum is obtained. This technique can identify the presence of functional groups and changes in their chemical environment and is widely used for qualitative and quantitative analysis of organic compounds, inorganic substances, polymers,

biomacromolecules, and even nanomaterials [1][2]. Modern FTIR extends beyond conventional transmission mode to incorporate multiple sampling techniques including attenuated total reflection (ATR-FTIR), enabling detection of solids, liquids, thin films, and even trace residues [3].

1.2 Thermal Analysis

Thermal analysis is a collective term for a range of analytical techniques that measure how the physical properties of a substance change with temperature under programmed temperature control. Through these techniques, key thermal performance parameters of materials can be obtained, including enthalpy change/heat flow, phase transition, glass transition, heat capacity, mass change, and coefficient of thermal expansion, etc [4]. Meanwhile, thermal analysis is widely used to study solid-phase reactions such as the analysis of degradation heat, decomposition products, and the determination of reaction kinetics parameters. Common specific techniques include differential scanning calorimetry (DSC) thermogravimetric analysis (TGA), thermomechanical analysis (TMA), etc [5]. This experiment will focus on using TGA and DSC for analysis.

2. Working principle

2.1 Infrared Radiation

In an FTIR system, broadband infrared light from the source first passes through a beam splitter that divides the beam into two paths: one directed to a fixed mirror and the other reflected to a movable moving mirror. The reflected beams recombine and interfere, generating an interferogram that varies with the moving mirror's displacement. This signal is captured by a detector. As infrared light of different frequencies produces constructive or destructive interference with varying path differences, the resulting interferogram essentially represents a time-domain signal of light intensity changes with path difference, containing all frequency components. Subsequently, Fourier Transform is applied to convert this time-domain signal into a frequency-domain spectrum, yielding an infrared absorption spectrum with wavenumber (cm^{-1}) on the x-axis and absorbance or transmittance on the y-axis [3].

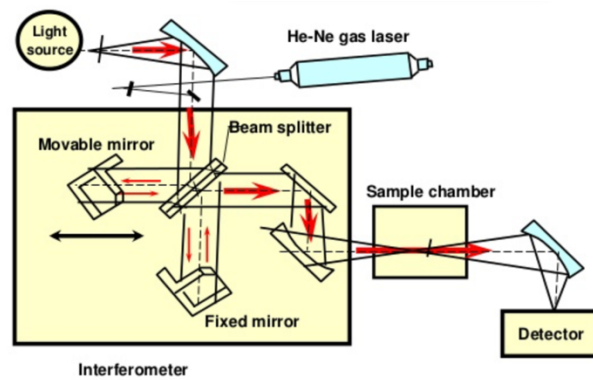


Figure 2. The working principle of FTIR

2.2 Thermal Analysis

2.2.1 TGA

Thermogravimetric Analysis (TGA) is a thermal analysis technique that measures the continuous change in sample mass with temperature or time under programmed temperature control [6]. The core principle is that during heating, substances often undergo mass changes due to physical changes (such as dehydration, vaporization) or chemical reactions (such as decomposition, oxidation). By monitoring these mass changes, the thermal stability, composition, decomposition kinetics, and reaction mechanisms of the material can be analyzed[6].

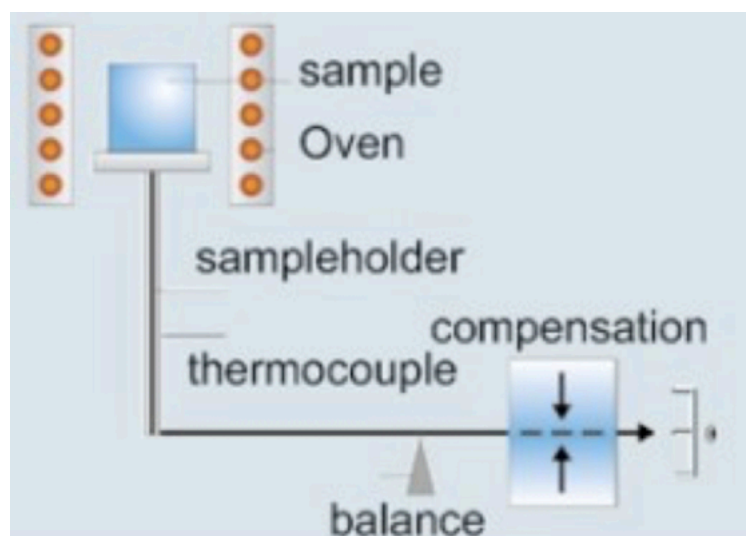


Figure 3. The components of the TGA instrument

2.2.2 DSC

Differential Scanning Calorimetry (DSC) is a method that directly measures the change in heat flow difference between a sample and an inert reference material as the temperature or time varies during a programmed temperature control process[7]. When a sample undergoes a physical transformation (such as melting, crystallization, or glass transition) or a chemical reaction, it absorbs or releases heat, resulting in a heat flow difference between the sample and the reference material. DSC can quantitatively characterize key thermodynamic and kinetic parameters of materials, such as melting enthalpy, crystallization enthalpy, glass transition temperature (T_g), specific heat capacity (C_p), reaction heat, and oxidation induction period, by precisely measuring this heat flow difference [8].

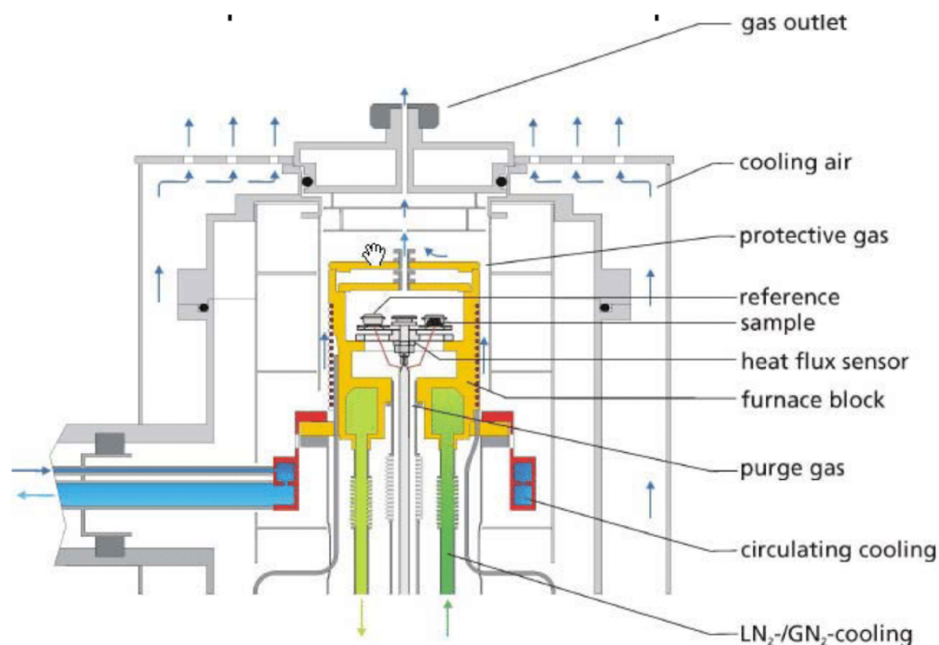


Figure 4. The components of the DSC instrument

3. Experimental instrument

3.1 Infrared Radiation

The instrument model: BRUKER TENSOR II

Main functions: It includes three modules: gas cell, thin film sample window and ATR, which can test the molecular vibration spectra of materials in solid, liquid and gas states.

Frequency range: 400 ~ 4000cm⁻¹



Figure 5. BRUKER TENSOR II

3.2 Thermal Analysis

This experiment utilized the SDT Q600 synchronous thermal analyzer to simultaneously conduct thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC). In a single experiment, the mass changes of the sample, and the heat flow information were simultaneously obtained to efficiently characterize the thermal behavior and structural stability of the material.



Figure 6. SDT Q600

4. Experimental procedure

4.1 Infrared Radiation

1. **Sample preparation:** Prepare samples according to testing requirements and grind solid samples into fine powder with a particle size below 200 mesh.
2. **Instrument status confirmation:** Turn on the FTIR spectrometer power and allow 30 minutes to 1 hour for preheating, ensuring the light source, detector, and interferometer are in a stable state.
3. **Instrument calibration:** Calibrate the instrument using known reference materials to ensure the accuracy of wavelength and intensity.
4. **Background scan:** Ensure the ATR crystal surface is clean, avoid placing samples for scanning, and preserve the background spectrum.
5. **Sample testing:** Take 5-10mg of fine powder or small sample pieces, place them on the surface of the ATR crystal, and apply appropriate pressure to ensure tight contact between the sample and the crystal.
6. **Data acquisition:** Perform sample scanning with parameters consistent with the background and save the spectra.
7. **Spectrum processing:** remove high frequency noise, eliminate background interference, and make the baseline flat.

4.2 Thermal Analysis

1. Weigh approximately 2mg of the sample using a precision balance and place it in an open alumina crucible.
2. Place the crucible containing the sample and an empty reference crucible separately in the sample and reference positions of the SDT Q600. Close the furnace.
3. Vacuum the furnace.
4. Introduce nitrogen gas to create an inert gas atmosphere.
5. In the software provided by TA Instruments, create a new method and set the temperature program: typically, increase the temperature from 50° C to 500° C at a rate of 10K/min.
6. After confirming that the parameters are correct, start the test program. The instrument will

automatically record the mass changes (TG) and thermal flow changes (DSC) of the sample during the heating process.

7. After the test is completed, wait for the furnace temperature to automatically cool down to below the safe temperature (or the specified safe temperature). Then, open the furnace cover to remove the sample. Shut off the gas source.

5. Analysis procedures

TGA

1. Export the TG curve (mass-temperature) and DTG curve (mass change rate-temperature) from the instrument. Check and correct the baseline drift.
2. Based on the TG and DTG curves, determine the temperature ranges for all mass losses (decrease steps) or increases (increase steps), with each range corresponding to a physical/chemical process.
3. Quantitatively extract for each stage: mass change Δm (%), starting temperature (T_{onset}), and peak temperature (T_{peak}).
4. Attribute the mass changes of each stage to specific processes (such as dehydration, decomposition, oxidation), and evaluate the thermal stability and composition of the material based on Δm and T_{onset} .

DSC

1. Perform baseline correction on the collected original DSC thermal flow curves to obtain a straight and stable baseline.
2. Identify the main thermal events based on the direction of heat flow and the shape of the peaks: downward heat-absorbing peaks are usually attributed to the melting process of the sample, upward heat-releasing peaks mostly correspond to crystallization or oxidation reactions, and the step-like transitions on the thermal flow curves can be determined as the glass transition (T_g).
3. Extract key quantitative parameters from the processed curves. The characteristic temperatures (such as the melting peak temperature T_m , the crystallization peak temperature T_c , and the midpoint of the T_g transition) can be directly read, while the heat enthalpy values of phase transitions or reactions (ΔH) can be calculated by integrating the corresponding peak areas and dividing by the sample mass.

6. Combination of FTIR and TG-DSC

The combined application of FTIR and TG-DSC—referred to as TG-DSC-FTIR or TG-FTIR—has become a standard configuration for studying the thermal behavior of complex materials. In this setup, volatile products generated during pyrolysis of samples under controlled heating are introduced into an FTIR spectrometer via a heating transfer line for real-time online analysis, enabling a tripartite characterization of "mass loss—thermal effect—gas composition." This strategy is particularly suitable for investigating polymer degradation mechanisms [9]. This demonstrates that FTIR focuses on extracting structural information at the molecular level, while TG-DSC emphasizes the quantitative analysis of macroscopic thermal behavior and physicochemical transformations. The synergistic application of these two techniques forms a crucial pillar in modern material characterization.

7. Analysis of Sample

A1 (Wang Yixuan)

1. Structural Analysis

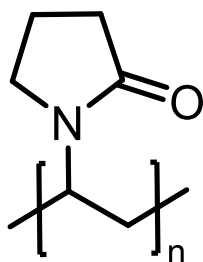


Figure 7. The structural formula of Polyvinylpyrrolidone

The main functional groups include N-H, C=O, C-N, and C-H in the ring, C-C chains.

2. Infrared analysis

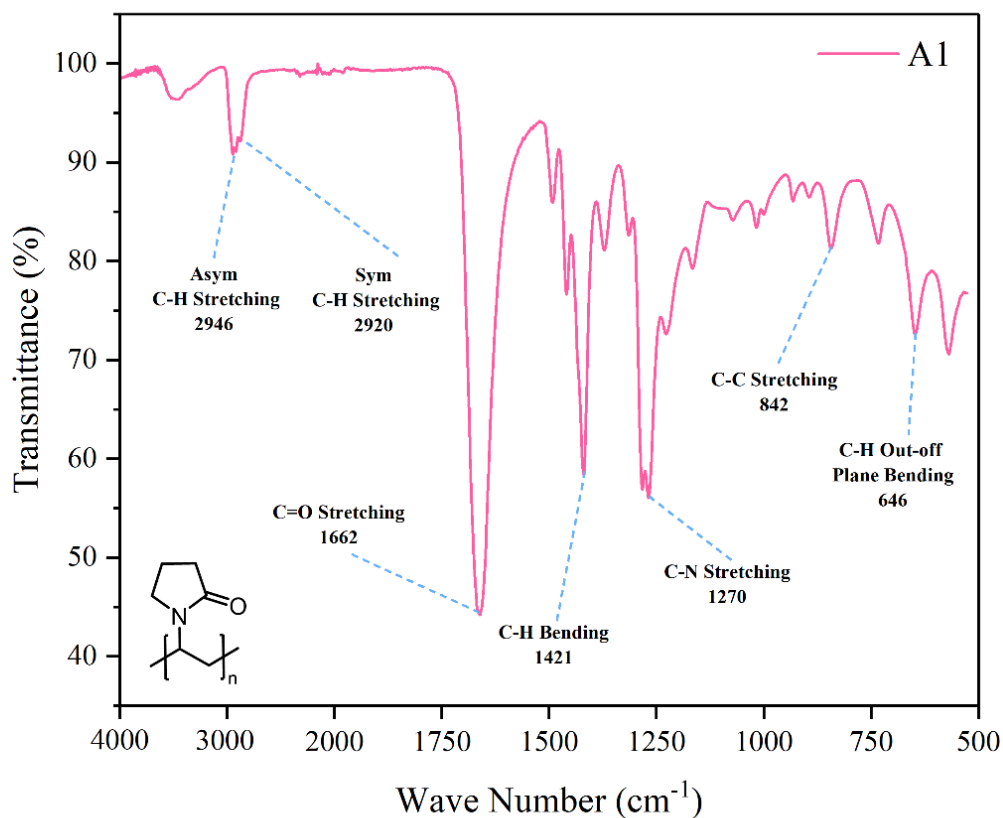


Figure 8. The experimental infrared spectra of PVP

Table 2. Infrared spectral analysis of PVP

Wave Number (cm^{-1})	Functional group	Vibrational mode
2946	C-H	Asymmetrical Stretching
2920	C-H	Symmetrical Stretching
1662	C=O	Stretching
1421	C-H	Bending
1270	C-N	Stretching
842	C-C	Stretching (Ring vibration)
646	C-H	Out of plane bending

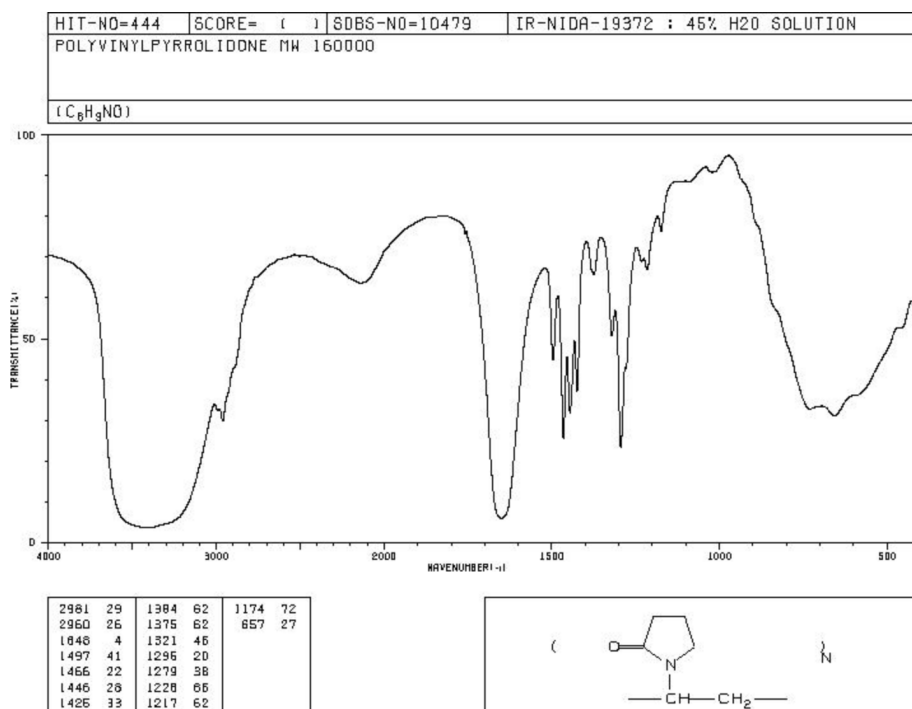


Figure 9. The standard infrared spectra of PVP [10]

By comparing our experimental spectra with the standard spectra of PVP, it is clearly observable that both spectra show C-H stretching vibration peaks in the range of 2950–2850 cm^{-1} , C=O stretching vibration peaks in the range of 1660–1640 cm^{-1} , C-H bending vibration peaks and C-N stretching vibration peaks in the range of 1500–1250 cm^{-1} , as well as C-C stretching vibration peaks and C-H out-of-plane bending vibration peaks in the range of 850–650 cm^{-1} .

However, there are also differences between the experimental spectrum and the standard spectrum. The most obvious difference is in the N-H stretching vibration peak in the $3400\text{--}3200\text{ cm}^{-1}$ range. The N-H stretching vibration peak in the standard spectrum has a larger width and is very distinct, while the N-H stretching vibration peak in the experimental spectrum is very small and difficult to observe. This might be due to the fact that the standard spectrum was tested with samples containing water or having a high adsorption capacity, while the experimental spectrum was tested with samples that were relatively dry and had less water content, resulting in the N-H stretching vibration peak in the spectrum being less prominent. Secondly, in the $850\text{--}650\text{ cm}^{-1}$ range, the C-C stretching vibration peak and the C-H out-of-plane bending vibration peak are presented. The standard spectrum shows a continuous broad peak, while the experimental spectrum shows a dispersed narrow peak. This might be related to the sample preparation method. Different sample preparation methods can lead to differences in the crystallinity of the samples, and the presence of the crystalline region can cause peak splitting and narrowing, while the peaks in the non-crystalline region are usually wider.

3. Thermal analysis

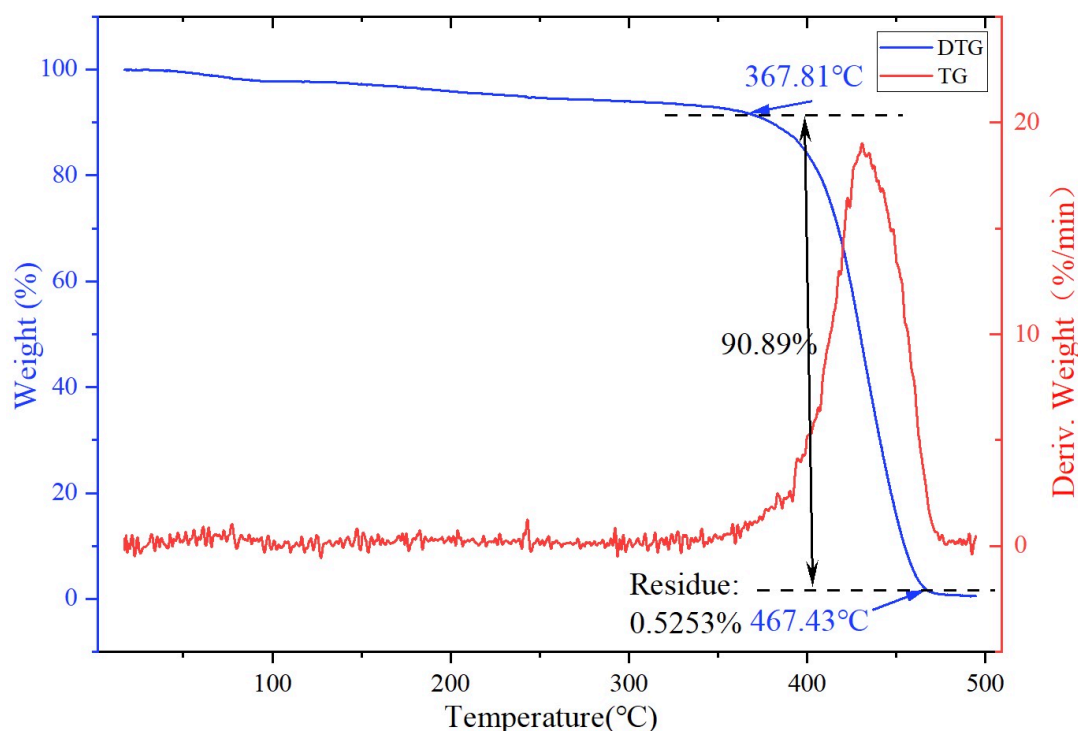


Figure 10. The TG and DTG curves of PVP

From the TG and DTG curves, this substance undergoes mass loss within two temperature ranges.

Firstly, from room temperature to 300°C, the mass loss is relatively small (less than 10% of the weight), the curve trend is stable. The main reason for the mass loss is the evaporation of physical adsorbed water or residual solvents. The insignificant curve change might be due to the low water content and the insignificant mass loss. The second stage is from 367.81°C to 467.43°C. During this stage, there is a rapid decrease in mass (a weight loss of 90.89%, with the remaining weight being only 0.5253%). It presents a steep mass loss plateau. The main reason for the mass loss in this range is the decomposition of PVP, including the ring-opening decomposition of the side chains of PVP and the breakage of the main chain.

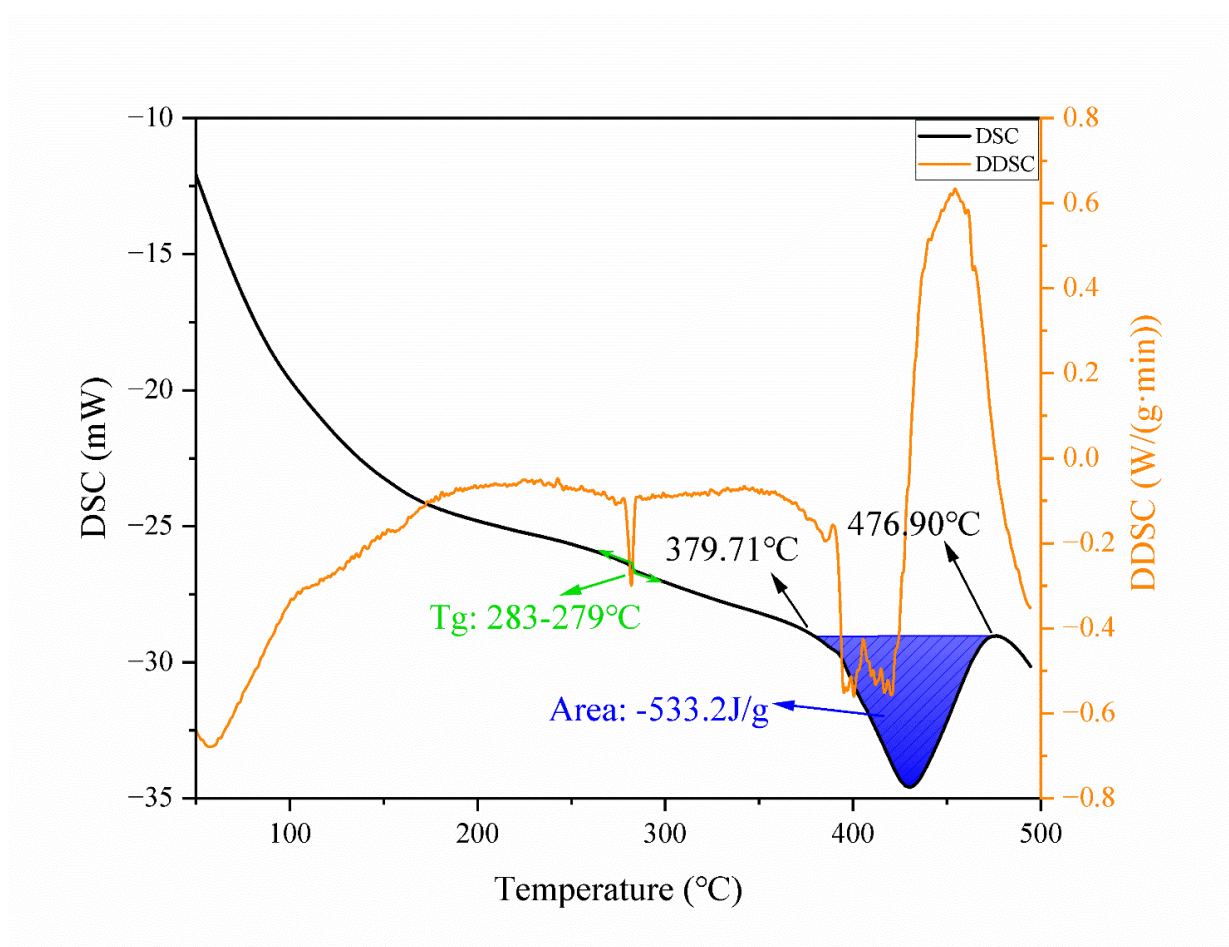


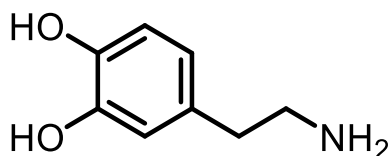
Figure 11. The DSC curve of PVP

By analyzing the DSC curve, it can be concluded that within the temperature range of 25°C to 200°C, the DSC curve is relatively stable and there are no obvious heat flow peaks, corresponding to the mass reduction in the TGA curve caused by evaporation of physical adsorbed water or residual solvents. Due to the low water content, the curve change is not significant. A small heat flow peak ($\Delta H \approx 0.5268$

J/g) appears between 279.36°C and 283.89°C (this is inferred to be the glass transition temperature (T_g) of PVP or the pre-decomposition stage before the main chain decomposition of PVP). Within the temperature range of 379.71°C to 476.90°C, the DSC curve significantly shifts downward, forming a wide valley peak ($\Delta H \approx 533.2$ J/g), which represents a strong exothermic process and the maximum exothermic peak is at 388.89°C, corresponding to the rapid mass decrease process caused by PVP decomposition (including the ring-opening decomposition of the side chains of PVP and the breakage of the main chain) in the range of 367.81°C to 467.43°C in the TGA curve. After 480°C, the heat flow gradually returns to the baseline, confirming that the remaining substances are only 0.5253% and there are no more reactive substances.

A2 (Che Xuanming)

1. Structure Analyse



The chemical structure of dopamine indicates that the molecule consists of a substituted benzene ring framework, with two ($-\text{OH}$) on the aromatic ring and a ($-\text{NH}_2$) on the side chain. These groups can readily form **intra- or intermolecular hydrogen bonding** in the system, thereby affecting its vibrational behaviour and spectral peak characteristics. The conjugated system of the benzene ring can induce a certain **conjugation effect**, causing the vibrations of functional groups connected to the aromatic ring to exhibit corresponding characteristic changes.

2. IR Analyse

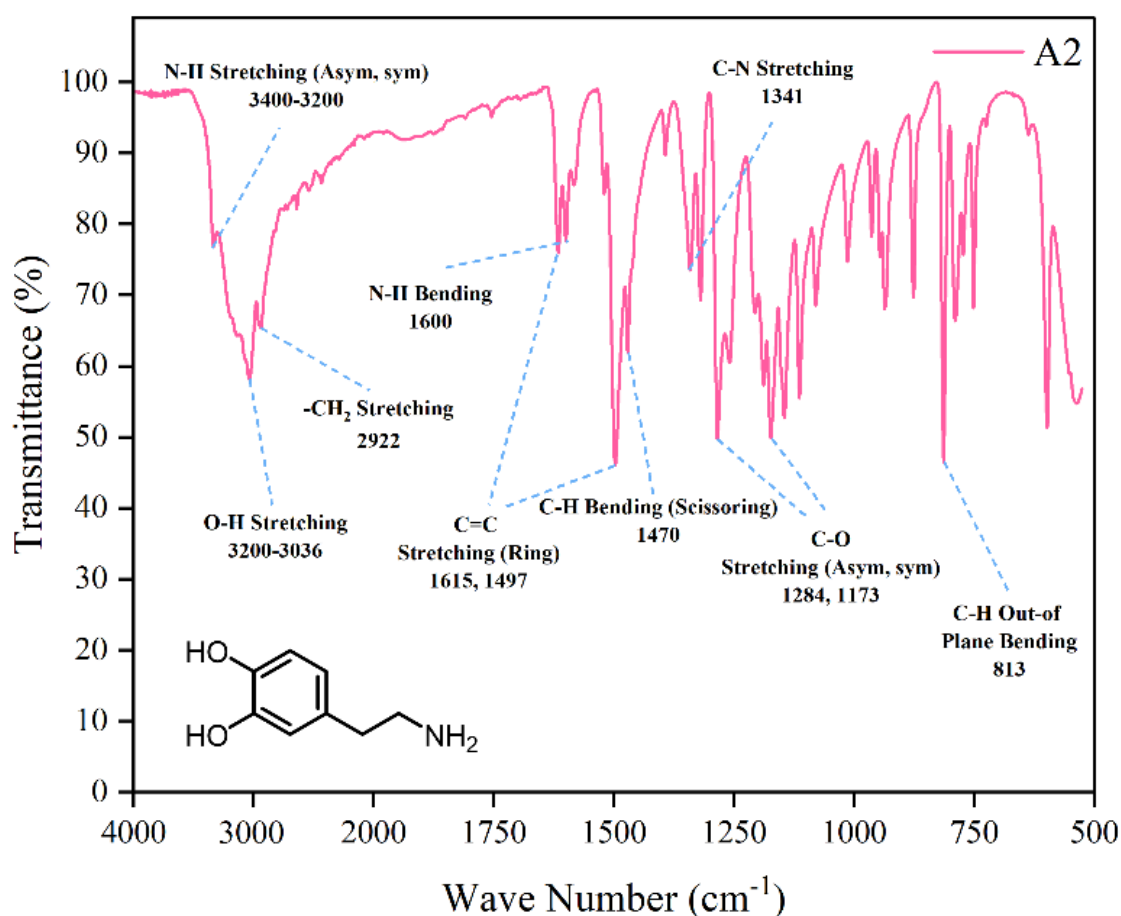


Figure 12. The experimental infrared spectra of dopamine.

Table 3. Infrared spectral analysis of dopamine.

Wavenumber (cm^{-1})	Functional group	Vibrational mode
3400-3200	N-H	Asym and sym Stretching
3200-3036	O-H	Stretching
2922	C-H (CH_2)	Stretching
1615	C=C	Stretching (Ring)
1600	N-H	Bending
1497	C=C	Stretching (Ring)
1470	C-H	Bending (Scissoring)
1341	C-N	Stretching
1284	C-O	Asymmetrical Stretching
1173	C-O	Symmetrical Stretching
813	C-H	Out of Plane Bending

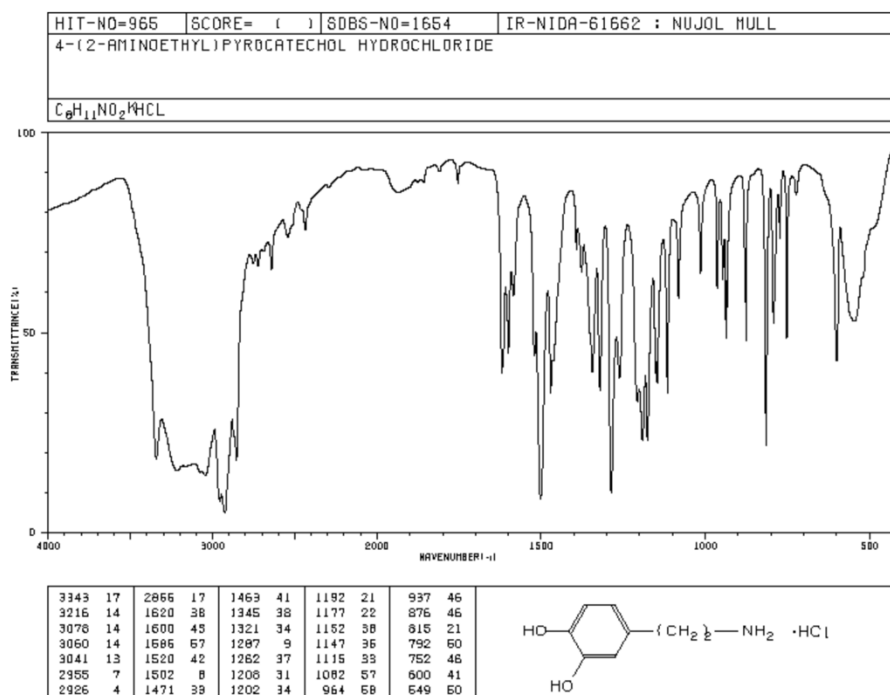


Figure 13. The standard infrared spectra of dopamine.

In the high-wavenumber region of 2800–3500 cm^{-1} . A broad absorption band with significant intensity is observed. This band mainly arises from the overlapping contributions of N–H (asymmetric/symmetric stretching) at 3400–3200 cm^{-1} and O–H stretching at 3200–3036 cm^{-1} . Since

both N–H and O–H are polar bonds and are highly sensitive to the hydrogen-bonding environment. In addition, the conjugation effect of the aromatic system leads to a relatively lower effective frequency for O–H stretching, making this absorption more prone to forming a tail near $\sim 3200\text{ cm}^{-1}$ or overlapping with the N–H region.

At 2922 cm^{-1} . The C–H stretching vibration of $-\text{CH}_2-$ is observed, indicating the presence of aliphatic methylene groups in the sample. In the subsequent mid-wavenumber region, the C–H scissoring (bending) vibration at 1470 cm^{-1} is also observed. Together, these two bands form a typical “stretching–bending” paired relationship.

In the skeletal vibration region of $1400\text{--}1650\text{ cm}^{-1}$. The aromatic C=C stretching vibrations (ring stretching) at 1615 cm^{-1} and 1497 cm^{-1} can be observed, which are key characteristic peaks of the aromatic ring structure. Meanwhile, 1600 cm^{-1} corresponds to N–H bending, and its wavenumber is close to that of the aromatic ring skeletal vibrations, which is consistent with the structural feature of the coexistence of an aromatic system and nitrogen-containing groups.

In the fingerprint region of $1300\text{--}400\text{ cm}^{-1}$. The spectrum provides direct evidence for nitrogen- and oxygen-containing functional groups: 1341 cm^{-1} is assigned to C–N stretching, while 1284 cm^{-1} and 1173 cm^{-1} correspond to asymmetric and symmetric C–O stretching, respectively. Because multiple substitution sites on the aromatic ring often introduce various vibrational modes and coupling, the fingerprint region shows numerous peaks and a complex spectral pattern.

At 813 cm^{-1} . C–H out-of-plane bending is observed, which is commonly used to characterize the substitution pattern of an aromatic ring. This band corresponds with the aforementioned aromatic C=C skeletal peaks at $1615/1497\text{ cm}^{-1}$.

Overall, the standard spectrum and the experimental spectrum are consistent to a certain extent, but differences in peak width and intensity are also present. These discrepancies may be caused by impurities in the experimental sample, moisture absorption from air, and inherent limitations of the instrument.

3. Thermal analysis

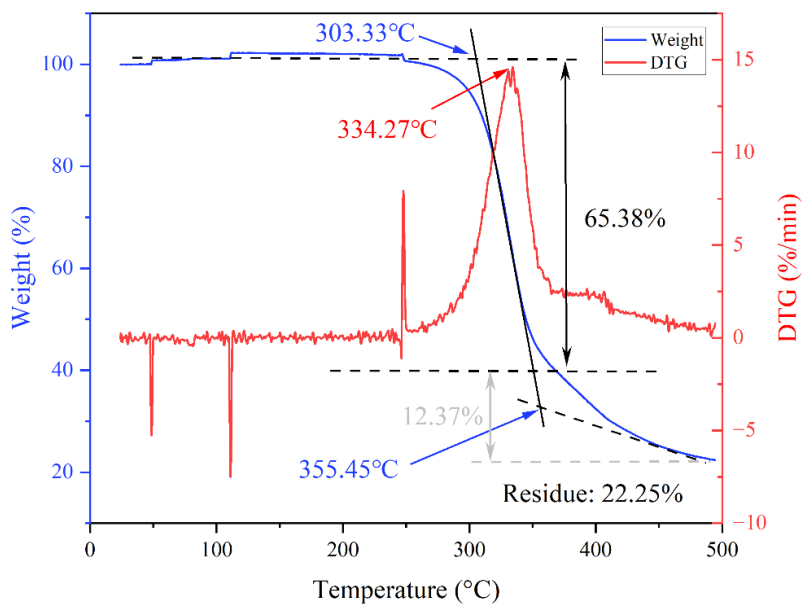


Figure 14. The TG and DTG curves of dopamine.

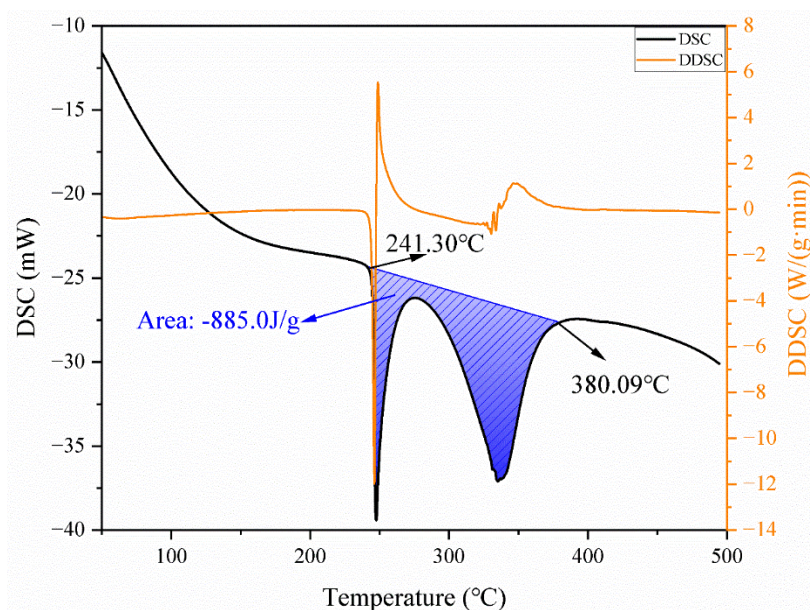


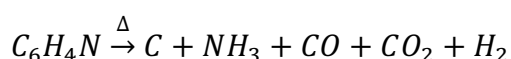
Figure 15. The DSC curves of dopamine.

1. The molten stage (241-247 °C). *LKT Labs* and other documents give dopamine hydrochloride a melting point at 241 °C [12]. The baseline offset at 241.30°C and the small peak at 245.46°C in the DSC-TGA graph correspond to sample melting. At this point, the TG curve remains almost unchanged, indicating no significant volatilization or weight loss. Since the peak in the DSC diagram absorbs heat downward, it is consistent with the heat absorbed by the melt.

2. The main decomposition stage (303-355 °C). As the temperature rose to 303 °C, the TG - DTG figure display quality drops rapidly, red DTG curves in 334.27 °C achieve maximum weight loss rate, blue curve has fallen from 100% to about 34.6%, 65.38% of weightlessness. This section corresponds to the first decomposition of the catechol ring and the chloride portion of dopamine hydrochloride, releasing small molecules such as HCl, H₂O, and CO₂, which is an extension of the catechol structure decomposition within the range of 210-330 °C in the literature [13].



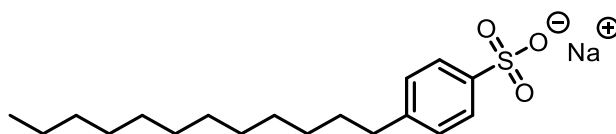
Subsequently, at 334-355°C, the TG curve further decreased by 12.37%, and the DTG curve showed a secondary peak. This stage corresponds to the decomposition of alkyl spacer groups and primary amine groups within the range of 330-465°C. A large amount of C-N and C-C bonds are cleaved to generate gases such as NH₃, H₂, CO and CO₂, accompanied by heat release.



3. The residue phase (355°C above) for more than 355 °C, the TG curve flattens, residual quality steady at around 22.25%, slightly higher than the 18.7% reported in the literature. The residues are mainly carbonised aromatic skeletons and inorganic salts, showing high thermal stability.

Analysis of Sample A3 (Zhang Yuchen)

1. Chemical structure analysis



Sodium Dodecylbenzenesulfonate (SDBS) is a highly significant anionic surfactant. Its infrared spectrum exhibits very characteristic peaks, clearly reflecting the three major structural units: the benzene ring, sulfonate group, and long-chain alkyl group.

2. Infrared analysis

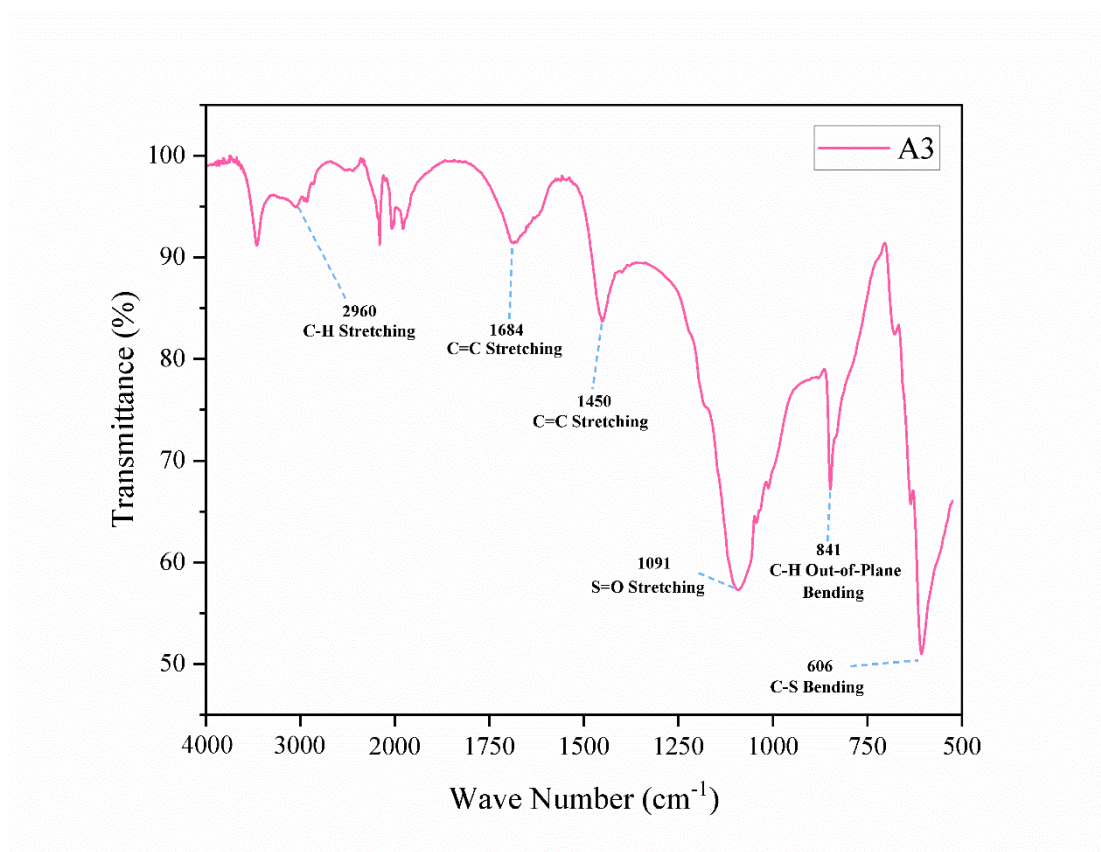


Figure 16. The experimental infrared spectra of SDBS.

Table 4. Infrared spectral analysis of SDBS.

Wavenumber (cm^{-1})	Functional group	Vibrational mode
3457	=C-H	Stretching
2960	CH ₂	Stretching
1684	C=C	Stretching in Benzene ring

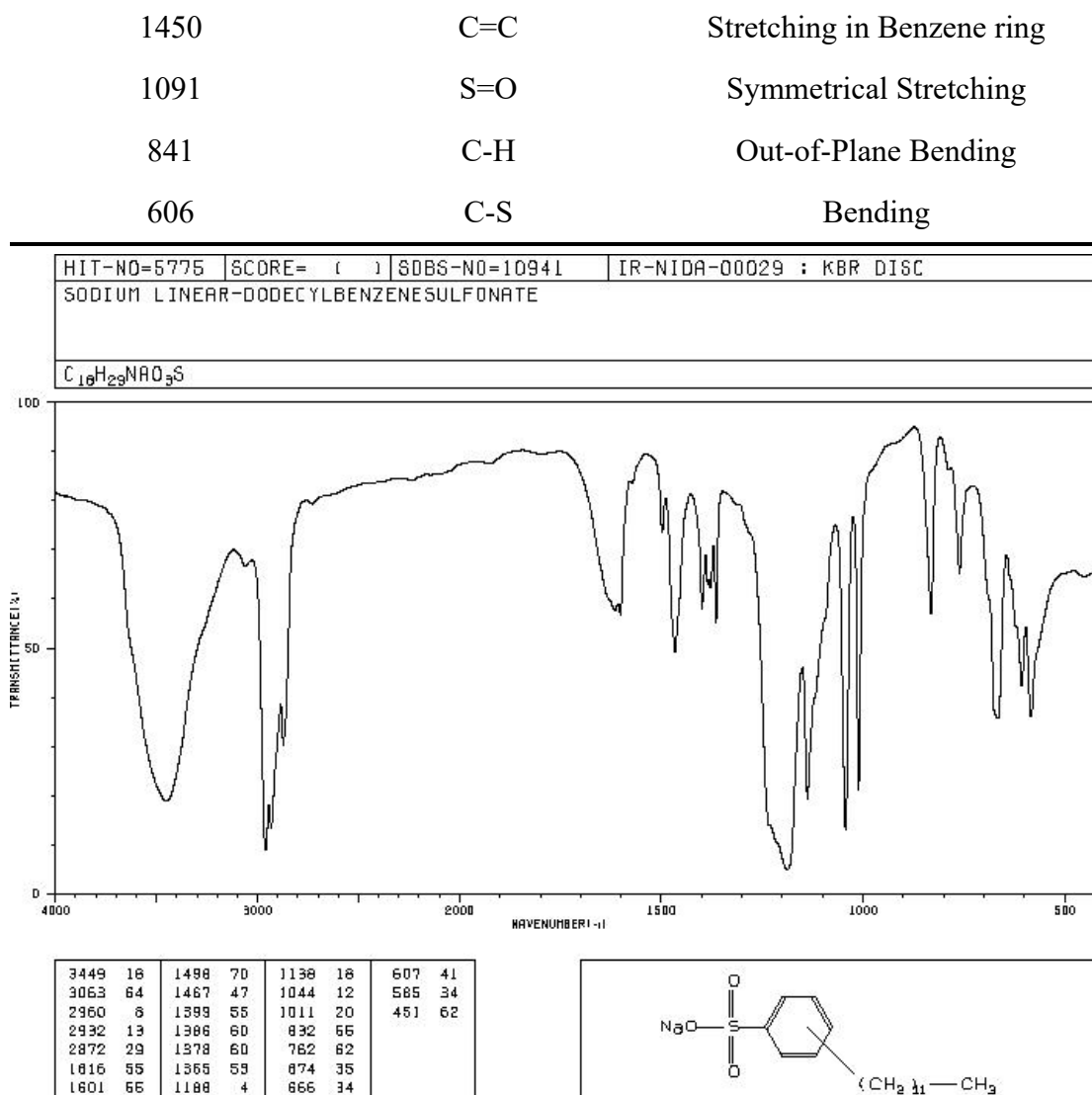


Figure 17. The standard infrared spectra of SDBS.

A comparison of the ATR-FTIR spectrum of sample A3 with the reference spectrum of sodium dodecylbenzenesulfonate (SDBS) reveals that A3 exhibits the characteristic fingerprint features of "long alkyl chain + substituted benzene ring + sulfonate group": an alkyl C–H stretching absorption appears at 2960 cm^{-1} ; overlapping absorptions attributed to aromatic ring skeleton vibrations and CH_2 –scissoring vibrations are observed around 1450 cm^{-1} ; a typical aromatic C–H out-of-plane bending peak of a substituted benzene ring appears at $\sim 841\text{ cm}^{-1}$; and a strong absorption band in the $1000\text{--}1200\text{ cm}^{-1}$ region, combined with a lower-wavenumber absorption around 606 cm^{-1} , collectively point to vibrations associated with the SO_3^- group (S=O, S–O/C–S), supporting that the main component of the sample is SDBS.

Meanwhile, some features commonly seen in the reference spectrum, such as the aromatic =C–H stretch around 3060 cm^{-1} and the doublet of aromatic ring skeleton vibrations near $\sim 1600\text{ cm}^{-1}$, are not clearly resolved in A3. Additionally, A3 shows a broad absorption around 3457 cm^{-1} and an atypical absorption in the $1650\text{--}1685\text{ cm}^{-1}$ region, which are more likely related to moisture absorption/hydration of the sample—leading to the broad O–H band and H–O–H bending vibration of water molecules—as well as to changes in relative intensity due to ATR, background subtraction, and spectral processing factors (resolution/smoothing) that may obscure fine peaks and merge spectral features. In the absence of distinct signals contradictory to the SDBS structure—such as a strong carbonyl band in the $1700\text{--}1750\text{ cm}^{-1}$ region—the overall combination of fingerprint region signals and characteristic peaks still allows the conclusion that A3 is consistent with the SDBS reference spectrum.

3. Thermal analysis

The mass and energy changes of the sample are mainly concentrated in the low-temperature range. In the initial stage, from room temperature to about 83°C , the mass remains essentially constant, indicating that the molecular framework as a whole is in a stable state. When the temperature rises from 83°C to 100°C , the first stage of rapid mass loss occurs, with a mass loss ratio of 6.486%, and the corresponding DTG peak temperature for the maximum mass loss rate is 87.99°C . Simultaneously, the recorded DSC curve exhibits a significant endothermic peak within this range, with a peak temperature of 84.53°C and an enthalpy change of approximately 53.48 J/g . The matching mass loss and strong endothermic signal jointly confirm that the essence of this process is the removal of adsorbed or bound water surrounding the hydrophilic sulfonate head groups in the sample. This process requires substantial energy to disrupt the hydrated coordination structure and hydrogen bonding network around the ionic head groups, subsequently leading to the initial densification of the molecular packing state.

As the temperature continues to rise from 145°C to 166.78°C , the sample undergoes a second stage of slower mass loss, with a mass loss ratio of 3.612%, and the corresponding secondary DTG peak appears at 149.9°C . Meanwhile, the DSC curve shows a persistent, broad, and gentle variation in the heat flow baseline over the range of 104.78°C to 176.63°C , forming a weak exothermic background

with a total enthalpy change of only 5.797 J/g. This broad-temperature, low-intensity thermal feature is not a sharp phase transition but rather points to a slow process of solid-phase structural relaxation and ordering. It reveals that after the main water loss, the microstructure of the sample does not immediately reach equilibrium. Instead, through mechanisms such as ion pair rearrangement and molecular conformational adjustments, it continuously absorbs a small amount of heat, gradually transitioning to a more stable anhydrous solid state.

It is noteworthy that when the temperature exceeds about 170°C and extends to the high-temperature range up to 500°C, the mass of the sample shows only a very slow decline, with a final residual mass as high as 88.15%. In sharp contrast, no significant endothermic or exothermic peaks are detected by the DSC curve over this entire broad high-temperature range. Pure SDBS usually undergoes vigorous exothermic or pyrolytic reactions due to alkyl chain cleavage and sulfonate group decomposition in the range of 600°C to 800°C [14], Therefore, when heated to 500°C, SDBS has not yet undergone thermal decomposition, and the residue mainly consists of undecomposed SDBS.

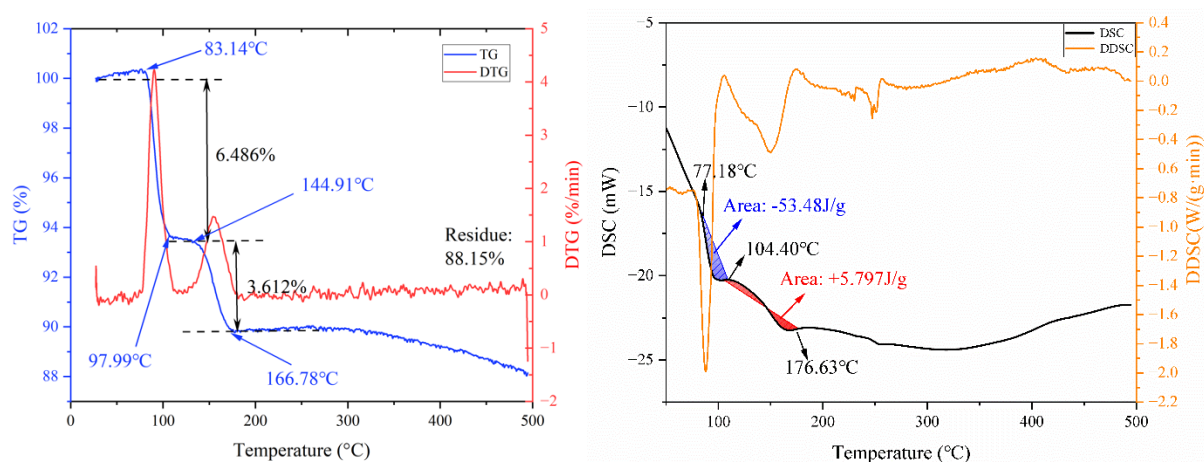
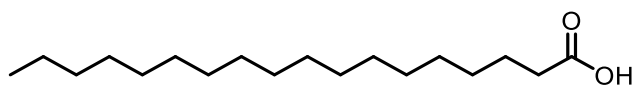


Figure 18. The T, DTG and DSC curves of dopamine.

A4 (Zheng Junhao)

1. Structure analysis



It is a linear, saturated long-chain fatty acid: an unbranched hydrocarbon (alkyl) chain terminated by a carboxylic acid group ($-\text{COOH}$).

2. Infrared analysis

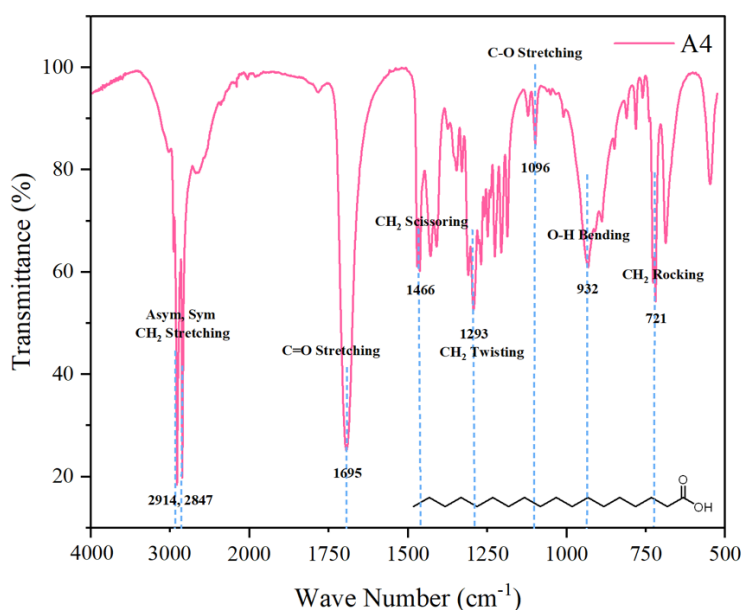


Figure 19. ATR-FTIR spectrum of stearic acid (sample A4)

Table 5. Infrared spectral analysis of stearic acid (sample A4)

Wavenumber (cm^{-1})	Functional group	Vibrational mode
2914	CH_2	Asymmetrical Stretching
2847	CH_2	Symmetrical Stretching
1695	C=O	Stretching
1466	CH_2	Scissoring
1293	CH_2	Twisting
1096	C-O	Stretching
932	O-H	Bending
721	CH_2	Rocking

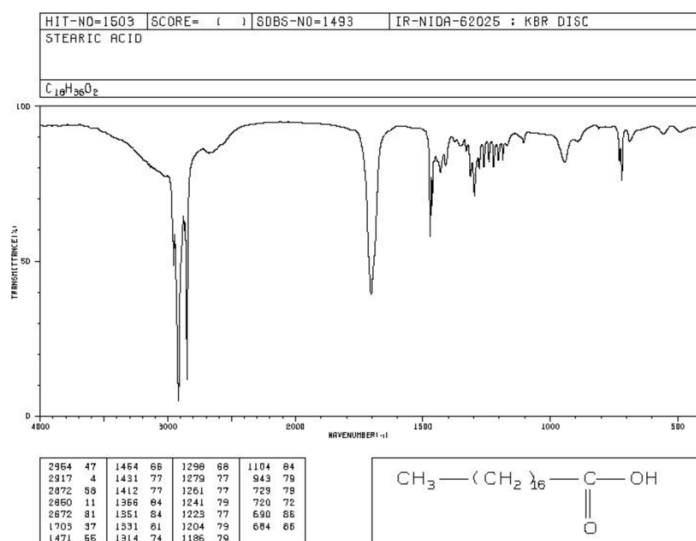


Figure 20. Standard FTIR spectrum of stearic acid (sample A4)

The ATR-FTIR spectrum of sample A4 matches well with the reference spectrum of stearic acid (KBr disc) in terms of the major diagnostic bands, despite expected differences in relative intensities and fine structures between ATR and transmission (KBr pellet) measurements. In the high-wavenumber region, A4 shows two intense bands at 2914 cm^{-1} and 2847 cm^{-1} , assigned to the asymmetric and symmetric stretching of methylene groups, which correspond closely to the reference bands at 2917 cm^{-1} and 2850 cm^{-1} . The carboxylic acid carbonyl band is observed at 1695 cm^{-1} in ATR, consistent with the reference $\nu(\text{C=O})$ at 1705 cm^{-1} . A slight downshift ($\approx 10\text{ cm}^{-1}$) is reasonable because ATR spectra often display small apparent shifts and different band shapes compared with KBr pellets due to the ATR penetration depth, refractive-index effects, and differences in sample crystallinity at the crystal interface.

In the fingerprint region, A4 retains the characteristic long-chain alkyl signatures of stearic acid. The reference shows CH_2 deformation bands near 1471 cm^{-1} ; in the experiment this appears as a single prominent band at 1466 cm^{-1} , consistent with CH_2 scissoring where the doublet can merge into one band in ATR or at lower resolution. The strong long-chain methylene rocking band in the reference at $729/720\text{ cm}^{-1}$ is reproduced by experiment at 721 cm^{-1} , confirming an extended linear $-(\text{CH}_2)_n-$ sequence. ATR also exhibits a band at 1096 cm^{-1} , corresponding to the reference 1104 cm^{-1} band in the C–O stretching vibration region of carboxylic acids, and a band at 932 cm^{-1} that aligns with the

reference 943 cm^{-1} band typically associated with carboxylic-acid O–H bending.

Overall, the agreement of the principal bands and their positions supports that sample A4 is consistent with stearic acid, and the observed discrepancies mainly originate from ATR vs KBr measurement mode and differences in spectral resolution and sample physical state rather than a different chemical identity.

3. Thermal analysis

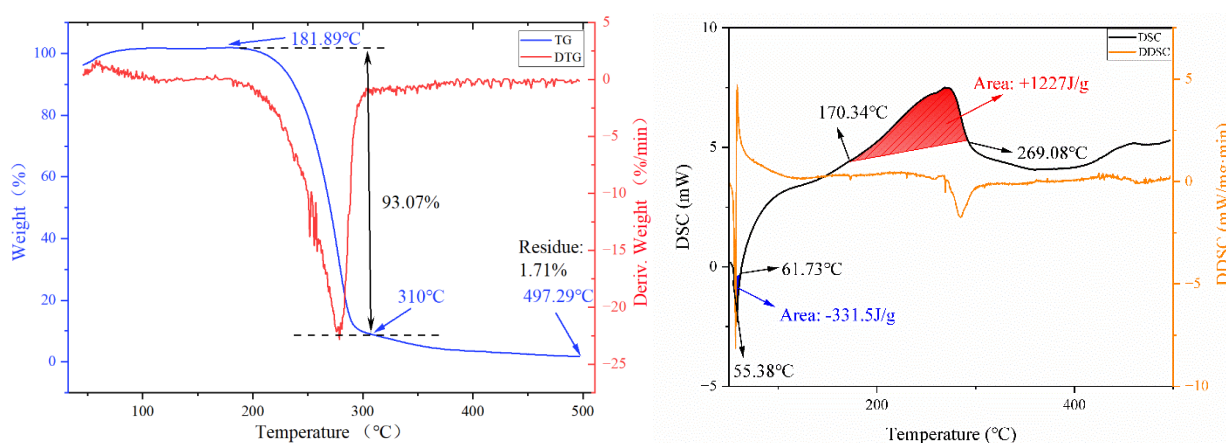


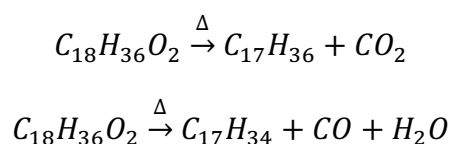
Figure 21. TG–DSC thermal analysis of stearic acid (sample A4) measured under N₂.

The simultaneous TG–DSC analysis of sample A4 under nitrogen indicates two main thermal stages. First, a distinct endothermic event is observed on the DSC curve between 55.38 °C and 61.73 °C (peak at 57.59 °C, $\Delta H \approx 331.5\text{ J g}^{-1}$), while the TG trace shows negligible mass change over the same interval. TG shows almost no weight loss in this temperature range. Combined with the endothermic peak of DSC, it is more consistent with solid-state phase transition/structural rearrangement (such as polymorphic solid-solid transformation) rather than volatilization or chemical decomposition; because if evaporation or decomposition occurs, it is usually accompanied by a significant decrease in sample mass. Such low-temperature solid-solid transformation has been reported in saturated long-chain fatty acid systems, and its transformation temperature can vary due to factors such as heating rate and sample thermal history [15].

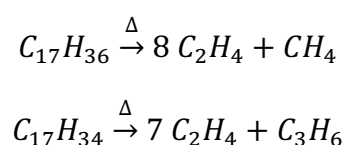
At higher temperatures, A4 undergoes the primary thermal degradation. The TG curve shows a rapid

mass loss from 181.89 °C to 310 °C, corresponding to an overall weight decrease of approximately 93.07%. This step is accompanied by a prominent thermal signal on the DSC curve with a peak at 269.08 °C (integrated heat $\approx 1227 \text{ J} \cdot \text{g}^{-1}$). Because the measurement was performed in N_2 , this process is attributed to pyrolytic decomposition accompanied by volatilization of low-molecular-weight products, rather than oxidative combustion [16]. Upon further heating to 497.29 °C, the residual mass is only 1.71%, indicating that the sample is predominantly organic and leaves minimal inorganic/ash residue. After consulting the references, sample A4 should have undergone the following reaction:

(1) Decarboxylation



(2) Secondary cracking

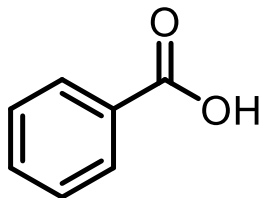


A literature review indicates that small apparent mass increases in TGA curves can occur due to instrumental artefacts. Specifically, the brief “apparent mass gain” ($\sim 101.88\%$) observed near 181.89 °C is most likely caused by buoyancy/baseline drift: as temperature rises, the furnace gas density decreases and the buoyancy term changes, which can produce a slight upward shift in the TG baseline rather than a real increase in sample mass [17].

Overall, the TG–DSC results support the ATR-FTIR identification: the IR evidence for a long $-(\text{CH}_2)_n-$ chain and a $-\text{COOH}$ group is consistent with the observed low-temperature solid-state transition and the subsequent extensive pyrolytic mass loss with very low residue, confirming that A4 behaves thermally as a long-chain saturated fatty acid (stearic acid).

A5 (Kong Xianglu)

1. Structural Analysis



This structure corresponds to benzoic acid; the chemical formula is $C_7H_6O_2$. It is formed by the connection of a benzene ring (C_6H_5-) and a carboxyl group ($-COOH$) and is the simplest aromatic carboxylic acid. The core functional group is the carboxyl group ($-COOH$), and it also contains a benzene ring (aromatic ring).

2. Infrared Analysis

From the chemical structural formula of benzoic acid, it can be concluded that its chemical bond types include C-O single bond, C=O double bond, O-H bond, C-H bond, and C=C double bond. Meanwhile, in infrared analysis, its vibration modes include bending vibration and stretching vibration, etc.

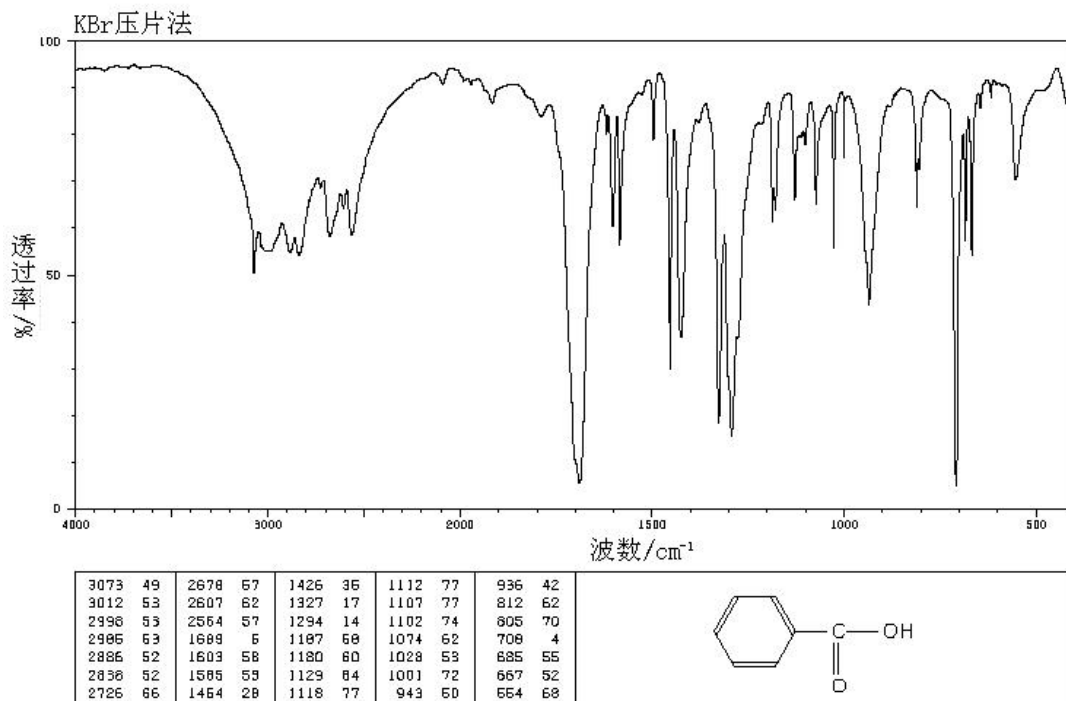


Figure 22. Standard infrared absorption spectrum of benzoic acid

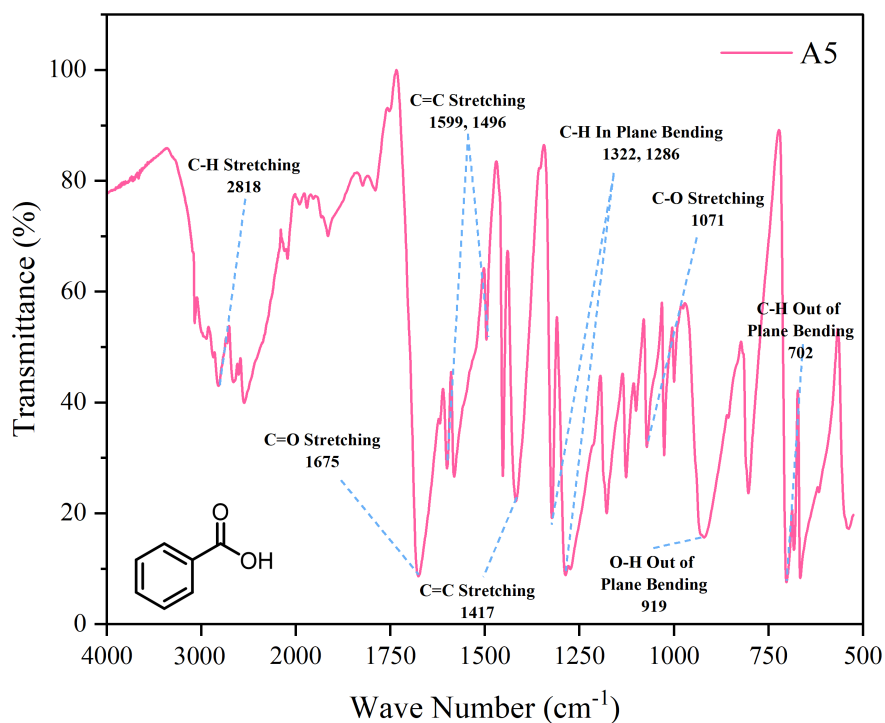


Figure 23. Infrared absorption spectrum of sample A5

Table 6. Infrared spectroscopy analysis results of Sample A5

Wave number (cm ⁻¹)	Vibrational mode
2818	C-H stretching
1675	C=O stretching
1599, 1496	C=C stretching
1322, 1286	O-H In-plane bending
1071	C-O stretching
919	O-H Out-of-plane bending
702	C-H Out-of-plane bending

From the infrared spectrum, the functional group region (4000–1300 cm⁻¹) exhibits typical features: the absorption peak near 2818 cm⁻¹ corresponds to the stretching vibration of C–H bonds on the aromatic ring; the peak at 1675 cm⁻¹ is attributed to the stretching vibration of the carboxylic acid carbonyl group (C=O), which shifts to a lower wavenumber compared to free carboxylic acid carbonyl groups (around 1700 cm⁻¹) due to intermolecular hydrogen bonding; the strong absorption peaks near

1599 cm^{-1} and 1500 cm^{-1} are assigned to the stretching vibrations of C=C bonds in the benzene ring skeleton, reflecting the electronic characteristics of the conjugated system of the benzene ring.

Entering the fingerprint region (1300–400 cm^{-1}), the absorption peaks at 1322 cm^{-1} and 1286 cm^{-1} correspond to the in-plane bending vibration of the O–H bond; the absorption at 1071 cm^{-1} is attributed to the stretching vibration of the C–O bond; the absorptions at 919 cm^{-1} and 702 cm^{-1} are associated with the out-of-plane bending vibration of the O–H bond and the out-of-plane bending vibration of the benzene ring C–H bond, respectively. Among them, the characteristic peak near 702 cm^{-1} is a typical marker of monosubstituted benzene ring structure (for monosubstituted benzene rings, C–H out-of-plane bending vibrations often appear in the ranges of 770–730 cm^{-1} and 710–690 cm^{-1} , and the substitution pattern can be confirmed by combining the peak shape).

Compare the characteristic peak positions, intensities, and functional group assignments of the sample's infrared spectrum with the standard infrared spectrum of benzoic acid one by one. The two are highly consistent in key functional groups (C=O, aromatic ring C=C), functional group coupling vibrations (such as O–H bending), and the pattern of characteristic peaks in the fingerprint region. Based on the analysis of the characteristic peaks in the infrared spectrum and the comparison with the standard spectrum, it can be determined that sample A5 is benzoic acid.

3. Thermal Analysis

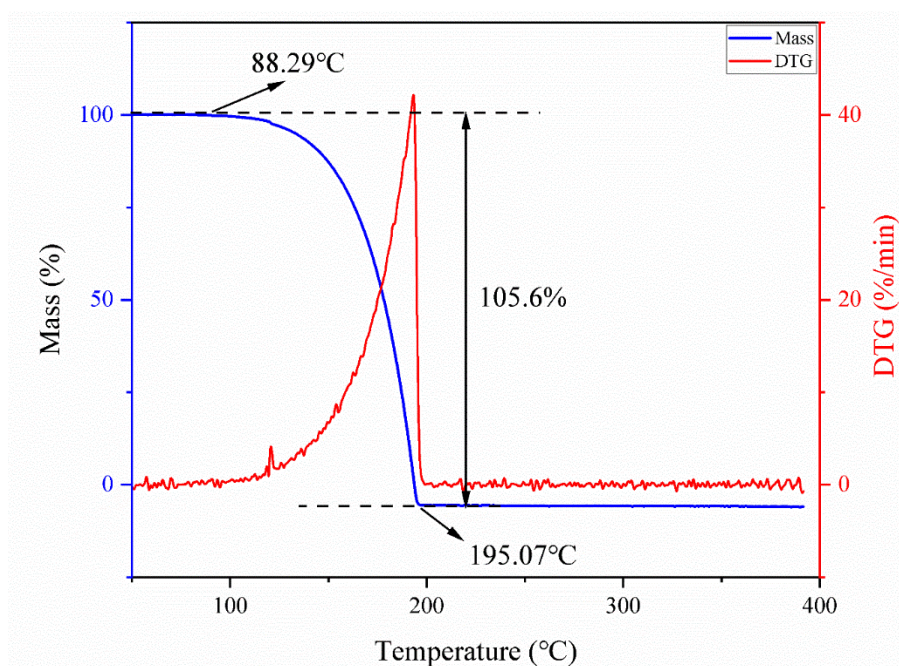


Figure 24. The TGA curve and DTG curve of sample A5

This figure shows that the TGA curve represents the weight percentage of the sample (left vertical axis, %), the DTG curve represents the derivative of the rate of weight change (right vertical axis, %/min), and the horizontal axis represents the temperature (0-400°C). The main features are as follows:

The sample begins to weight loss at 88.29°C (the inflection point of the TGA curve), and the DTG curve reaches its peak at 195.07°C (the maximum weight loss rate), indicating that the weight loss rate is very fast at this stage. Considering the nature of benzoic acid (which sublimates about 100 °C), this temperature range (88-195°C) can be presumed to be the weight loss caused by the volatilization of benzoic acid molecules. However, the solid-liquid transition does not cause a change in weight. Therefore, whether the solid-liquid transition occurs before this needs to be discussed in the DSC curve below. After this, benzoic acid should be completely sublimated, and the residual amount should approach zero.

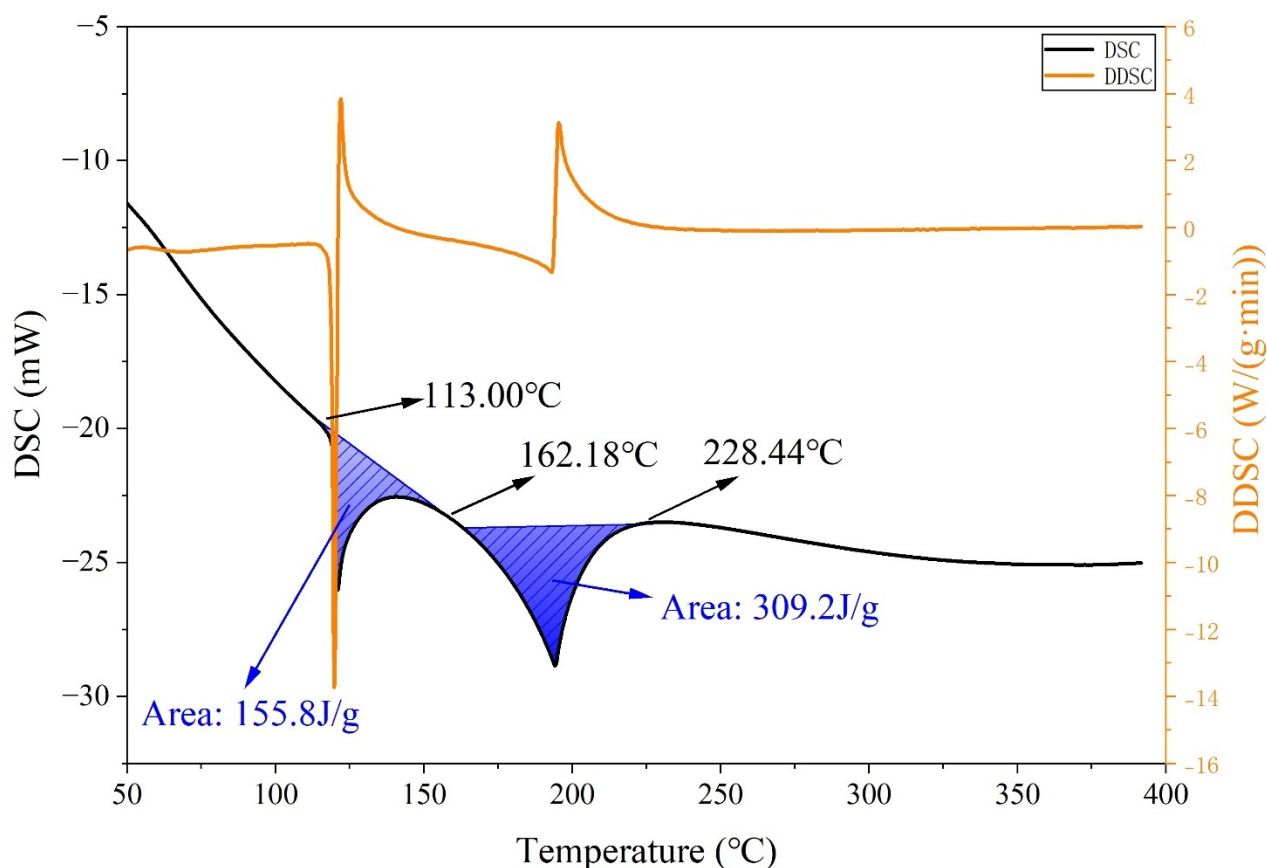


Figure 25. DSC diagram of sample A5

This figure presents the DSC curve with heat flow (left vertical axis, W/g) and the DDSC curve representing the derivative of heat flow (right vertical axis, W/g·min). The horizontal axis denotes temperature (50–400°C). Key thermal effect peaks are analyzed as follows:

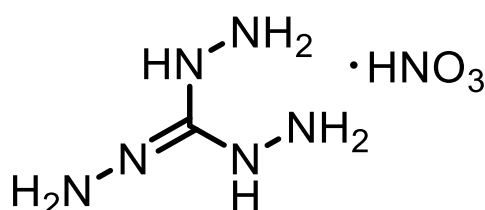
Melting Endothermic Peak (Low-Temperature Region): An endothermic peak appears at 120.89°C (with the DSC curve concave downward), corresponding to an enthalpy change of 155.8 J/g (peak area integral). This temperature range closely matches the standard melting point of benzoic acid (122.4°C), with minor deviations attributed to differences in sample particle size and heating rate. The enthalpy change value (155.8 J/g) is largely consistent with the reported melting enthalpy of benzoic acid (≈ 147 J/g) in the literature, confirming that the sample undergoes a solid-liquid phase transition (melting process) at this stage.

Evaporation endothermic peak (Medium-High Temperature Region): A weak endothermic shoulder peak occurred at 162.03–228.44°C, combined with the TGA weight loss initiation temperature (168.63°C), indicating that benzoic acid vaporized during this process. The endothermic peak at 194.19°C corresponds to an enthalpy change of 309.2 J/g, which is the evaporation enthalpy corresponding to the vaporization of benzoic acid. The DSC curve was stable within the range of 228.44–400°C, with no obvious thermal effect, further confirming that the sample mainly evaporated at high temperatures without vigorous decomposition.

A6 (Han Yujia)

1. Structural analysis

Tris (amino guanidinium) nitrate (TAGN) is an energetic compound, whose molecular structure is composed of a Tri aminoguanidine cation and a nitrate anion. The structure contains multiple N–H bonds (from the amino and guanidine groups), C–N bonds, N–N bonds, and nitrate (NO_3^-) groups. These functional groups have rich vibration modes and high structural symmetry, and clear peaks can be obtained in the infrared spectrum.



2. Infrared analysis

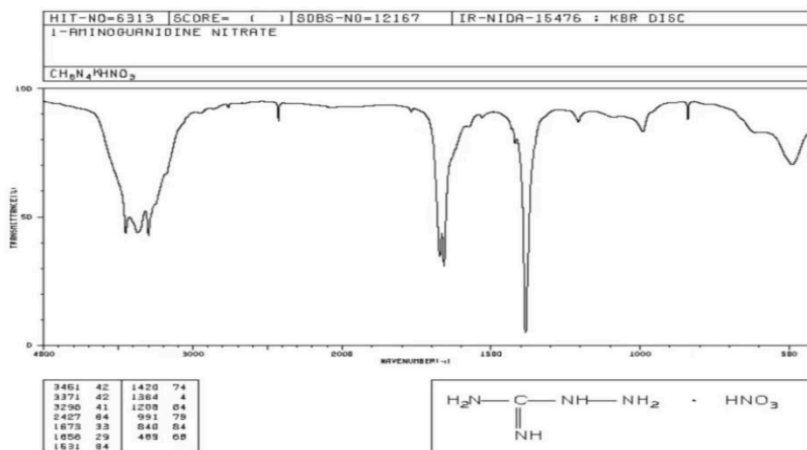


Figure 26. Standard infrared absorption spectra of TAGN

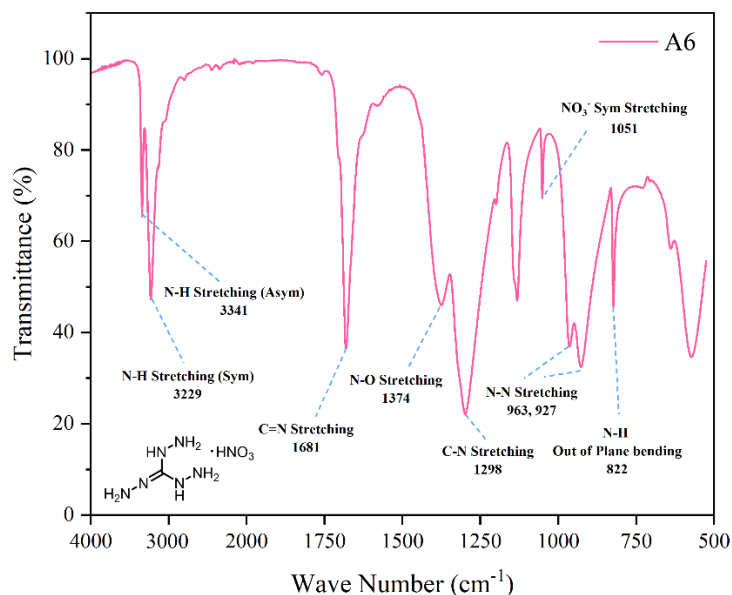


Figure 27 Infrared absorption spectra of TAGN [18]

Two sets of strong absorption peaks in the high wavenumber region (3341 cm^{-1} , 3229 cm^{-1}) correspond to the asymmetric and symmetric stretching vibrations of the secondary amine group ($-\text{NH}_2$) — this "bimodal feature" not only reflects the presence of multiple $-\text{NH}_2$ groups in the triaminoguanidine cation, but also, due to the vibration coupling between the groups, demonstrates the spatial distribution characteristics of the amino group on the guanidine skeleton. The strong absorption peak at 1681 cm^{-1} corresponds to the $\text{C}=\text{N}$ stretching vibration of the guanidine group, with its peak position falling within the typical range of $1620\sim 1690\text{ cm}^{-1}$, and the peak shape is sharp, consistent with the stabilizing effect of the guanidine conjugated system on the double bond vibration, further confirming the guanidine skeleton structure of the cation.

The absorption peaks at 1374 cm^{-1} and 1298 cm^{-1} in the mid-wave region correspond to the $\text{C}-\text{N}$ single bond stretching vibration of the guanidine skeleton, matching the connection mode of the $\text{C}-\text{N}$ bond in the triaminoguanidine cation; the characteristic peak at 1051 cm^{-1} belongs to the symmetric stretching vibration of the nitrate anion (NO_3^-), and its strong absorption intensity not only conforms to the infrared active characteristics of the planar trigonal structure of the nitrate anion, but also directly proves the presence of anions in the sample.

The infrared spectra of the experimental samples were compared with the standard infrared spectrum of triaminoguanidine nitrate (TAGN). The characteristic absorption peak positions, relative intensities and peak shapes of the main functional groups in the sample infrared spectra were all highly consistent

with those of the standard spectrum. This further confirmed that the measured samples were triaminoguanidine nitrate.

The unmarked peaks are believed to be caused by damage of water or impurities during the experimental operation. The results are shown in the table below:

Table 7. Infrared spectral analysis of TAGN

Wave number (cm^{-1})	Functional group	Vibrational mode
3341	N-H	Asymmetrical stretching
3229	N-H	Symmetric stretching
1681	C=N	Stretching
1374	N-O	Stretching
1298	C-N	Stretching
1051	NO ₃ ⁻	Bending
963/927	N-N	Stretching
822	N-H	Out of plane bending

3. Thermal Analysis

3.1 Analysis of TG and DTG curve

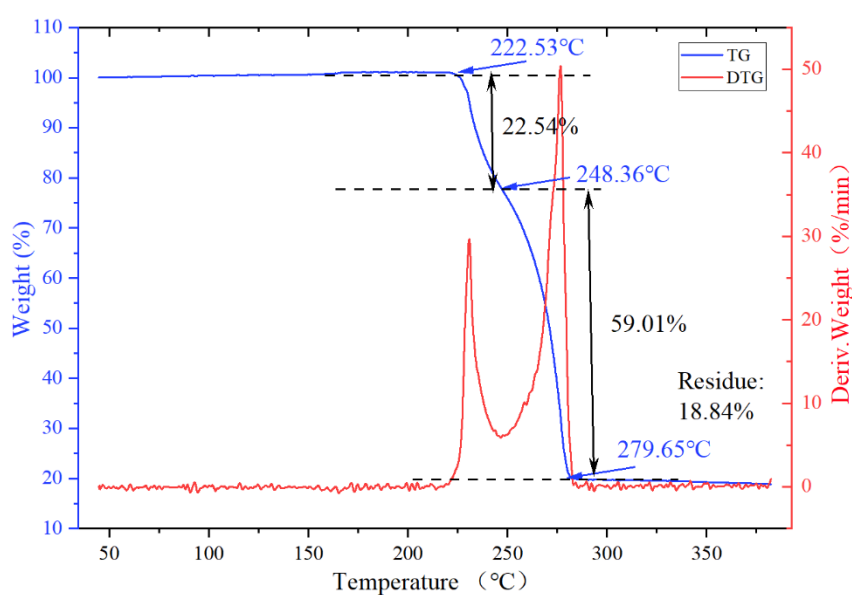
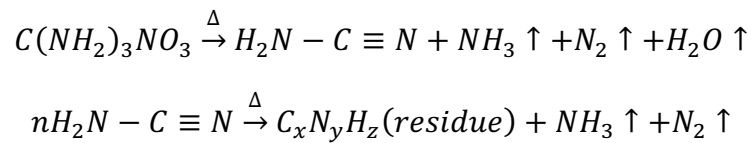


Figure 28. TG figure of sample 6 (black) and DTG figure of sample 6 (blue)

From the TGA curve in the graph (includes the solid line representing the weight change with temperature and the derivative weight curve), it can be directly observed that the initial mass of the sample is 3.0430 mg as indicated in the figure: Before 222.53°C, the curve remains basically stable, and no significant mass change occurs in the sample within this temperature range; When the temperature rises to 222.53°C, the curve begins to drop rapidly, and the weight loss process basically ends at 279.65°C, with the final residual mass shown in the figure being 0.5730 mg. The calculated total weight loss rate is approximately $\frac{3.0430-0.5730}{3.0430} = 81.2\%$, and the residual rate is $\frac{0.5730}{3.0430} = 18.8\%$.

Meanwhile, the sharp peaks in the derivative weight curve at 223.17°C and 263.43°C in the figure correspond to these two temperature points being the stages with the fastest weight loss rate, which directly reflects that this thermal decomposition is not a single-stage reaction, but a process involving multiple steps of bond rupture and product release. It is speculated that the multi-step reactions that occurred are as follows:



3.2 Analyze of DSC and DTG curve

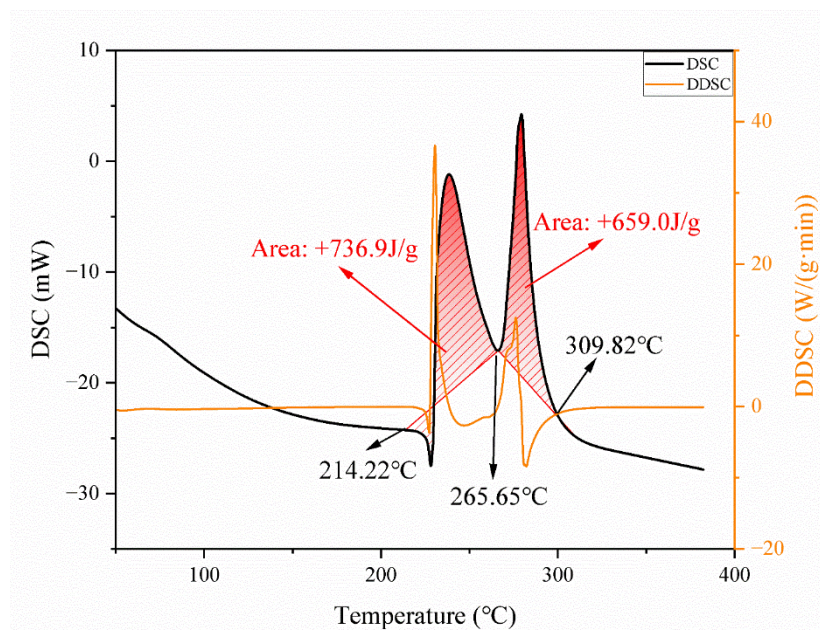


Figure 28. DSC figure of sample 6

From the DSC curve in the above figure, it can be observed that the thermal flow variation is in synchronous correlation with the weight loss process of TGA: The first peak of DSC occurs around 214.22°C, at this time the sample weight has not yet undergone significant fluctuations in the TGA curve, indicating that this peak is the endothermic process of the triaminoguanidine nitrate undergoing crystal transformation and removing weakly bound groups during the heating process, without accompanying significant mass changes.

The second peak of DSC (at 223.92°C) corresponds to the rapid weight loss stage in the TGA curve (the sharp weight drop after 222.53°C). This strong exothermic peak is the core thermal signal of the intramolecular oxidation-reduction decomposition reaction of the triaminoguanidine nitrate molecule. The corresponding reaction enthalpy is 739.8 J/g. The subsequent weak peak at 265.65°C corresponds to the stage after the TGA curve approaches the weight loss endpoint at 279.65°C. At this point, the sample has formed residual solid, and this peak is only the thermal effect of the further slight decomposition of the residual product, with an enthalpy of 269.0 J/g.

Based on the previous infrared spectroscopy analysis of functional groups and the multi-step weight loss and residual characteristics of TGA, the thermal behavior matching degree of the two further confirmed that this sample is Tris (amino guanidinium) nitrate.

8. Discussion

8.1 Infrared spectrum

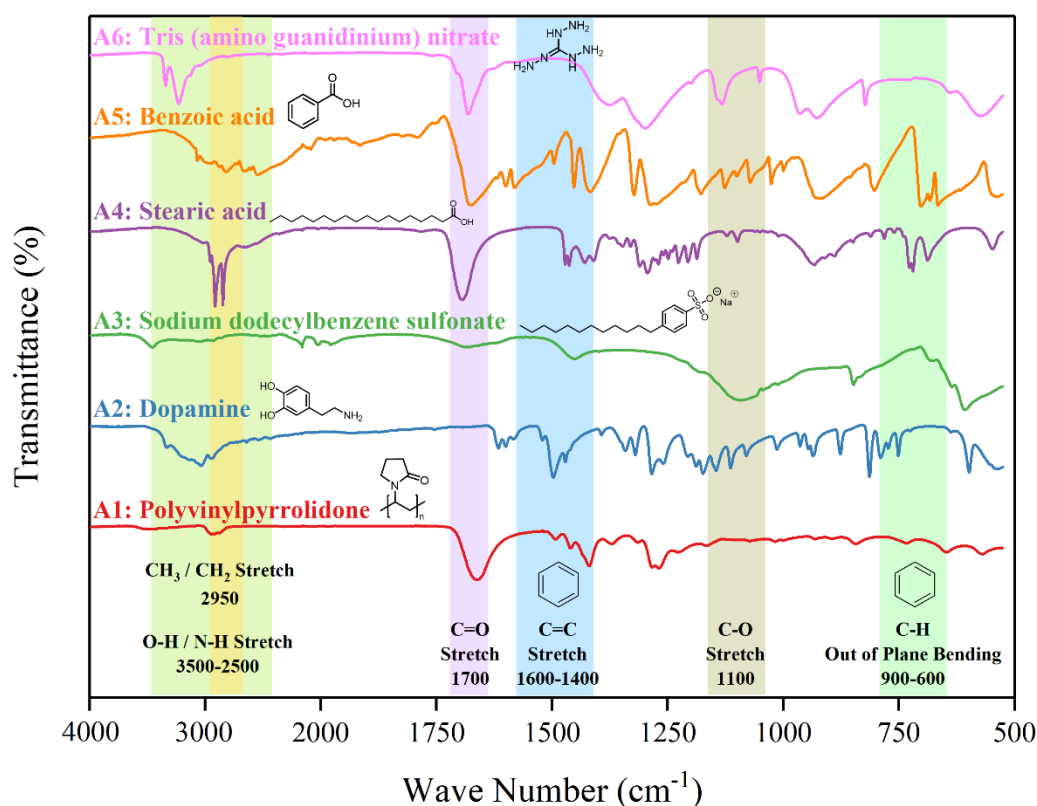


Figure 29. Overlay of the IR spectra for samples A1–A6

Figure 28 summarizes the IR spectra of samples A1–A6. While all spectra show C–H stretching features around 3000 cm^{-1} to varying extents (aliphatic and/or aromatic), clear differentiation arises from the diagnostic regions near $\sim 1700\text{ cm}^{-1}$ (C=O), $1600\text{--}1400\text{ cm}^{-1}$ (aromatic skeletal vibrations and/or deformation bands), and $1200\text{--}1000\text{ cm}^{-1}$ (C–O/C–N or heteroatom-related stretches such as S=O). Sample A4 (stearic acid) is dominated by long-chain aliphatic $\text{--}(\text{CH}_2)_n\text{--}$ signatures, including intense CH_2 stretching bands at $2914/2847\text{ cm}^{-1}$ and a characteristic CH_2 rocking band near 721 cm^{-1} , together with a carboxylic-acid carbonyl absorption around $\sim 1695\text{ cm}^{-1}$. Sample A5 (benzoic acid) also contains a --COOH group; however, due to aromatic conjugation and distinct solid-state packing, its carbonyl position and fingerprint-region profile follow the typical pattern of an aromatic carboxylic acid, and the 702 cm^{-1} region shows diagnostic aromatic C–H out-of-plane bending bands. Sample A1 (polyvinylpyrrolidone) exhibits a lactam-type C=O band in the $\sim 1662\text{ cm}^{-1}$, accompanied by C–N related features near the 1270 cm^{-1} . Sample A2 (dopamine) combines aromatic ring vibrations ($\sim 1600\text{ cm}^{-1}$) with broad N–H stretching contributions. Sample A3 (sodium dodecylbenzene sulfonate) is

distinguished by strong sulfonate-related absorptions in the 1091 cm⁻¹ region (S=O vibrations), in addition to aromatic features. Sample A6 (amino guanidinium nitrate) is characterized by prominent N–H stretching bands and nitrate-related absorptions, giving a spectral profile typical of an ionic salt and clearly different from neutral organic acids.

Importantly, variations in peak position and band shape for the same functional group across different molecules are expected and normal. Such differences mainly originate from (i) hydrogen-bond strength and aggregation state (carboxylic acids often form dimers, producing very broad O–H bands that extend to lower wavenumbers), (ii) electronic effects and conjugation (aromatic conjugation can shift the C=O frequency relative to aliphatic acids), (iii) polymorphism/crystallinity and molecular packing (different local environments in the solid state), and (iv) measurement mode and contact conditions (ATR vs transmission can alter relative intensities and band shapes). For example, although both A5 (benzoic acid) and A4 (stearic acid) contain –COOH groups and therefore show hydroxyl-related broad absorption and a carbonyl band, differences in the exact band shape and wavenumber are commonly caused by distinct hydrogen-bond networks, conjugation, and packing rather than any change in functional-group identity. Consequently, reliable identification should be based on the combined agreement of key band positions, fingerprint-region patterns, and overall spectral profiles.

8.1.1 Why do the frequencies of different molecular vibrations differ?

The vibration frequency of a molecule is determined by its force constant k and the masses of the atoms involved in the vibration μ , it conforms to the following formula:

$$\nu = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

In equation 2, m_1, m_2 are the atomic masses at both ends of the bond.

It shows in equation 1 that the larger the force constant k is, or the smaller the mass of the vibrating atoms μ is, the higher the vibration frequency will be, the peak position will appear in the high wavenumber region.

For example, the stretching frequency of the C-H bond is higher than the bending vibration frequency of the C-H bond. This is because the stretching vibration changes the bond's length, the force constant is larger; while the bending vibration only changes the bond angle, the force constant is smaller. This is also the reason why the stretching vibration frequency is generally higher than the bending vibration frequency. For the C-H bond and the C-C bond, the mass of the vibrating atoms of C-H bond is $\mu \approx 0.92$, the mass of the vibrating atoms of C-C bond is $\mu \approx 6$. The mass of the vibrating atoms of C-H bond is smaller, so it has a higher vibration frequency.

However, since generally the force constant difference of the bond is much greater than the difference in atomic masses involved in the vibration, in the determination of the molecular vibration frequency, the influence of the force constant k plays a dominant role.

8.1.2. Why does the experimental spectra obtained from infrared spectroscopy differ from the standard spectra?

The infrared spectra obtained from the experiments usually differ from the standard infrared spectra. This may be the result of the combined effects of various factors such as the sample state and experimental conditions. The standard spectra are typically in an "ideal state" (such as pure substances, dilute solutions, specific crystal forms), while the experimental samples may have additional infrared absorption peaks due to residual solvents, water, etc. during sample processing, such as O-H vibrations. Secondly, the experimental instruments used to create the standard spectra may have parameter differences from those used in the actual experiments. For example, the standard spectra are measured using high-resolution instruments (with sharp peaks and good separation), while the instruments used in the experiments have low resolution and result in wider peaks and severe overlap. Additionally, different sample preparation methods can also lead to differences in infrared absorption peaks. For instance, using the liquid film method to prepare samples may introduce impurities, resulting in interference peaks.

8.2 Thermal analysis

8.2.1 What should we primarily extract from TG, DTG, DSC and DDSC, and why are they interpreted together?

TG directly records the mass change of a sample with temperature, so it is mainly used to identify mass-loss steps, their onset/end temperatures, and the residual mass. DTG is the first derivative of TG and therefore highlights the rate of mass change; its peak temperature corresponds to the maximum mass-loss rate, which helps resolve overlapping or multi-step decomposition events that may look like a single step in TG. DSC measures the heat flow associated with thermal events, allowing identification of endothermic/exothermic transitions such as solid–solid transitions, melting, crystallization, and decomposition enthalpy changes. DDSC (the derivative of the DSC signal) further emphasizes subtle inflection points and overlapping thermal events, improving the determination of characteristic temperatures when the DSC baseline is sloped or peaks are broad [19]. Interpreting these curves together provides a consistent “mass–heat” evidence chain: events showing a DSC peak without a corresponding TG change are typically physical transitions, whereas DSC features that coincide with TG/DTG mass loss indicate chemical decomposition or volatilization, leading to a more reliable assignment of thermal processes than any single curve alone [19].

8.2.2 Why can the TG curve show an “apparent mass gain” or an abnormal mass-loss value (e.g., >100%) in some temperature ranges?

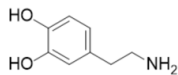
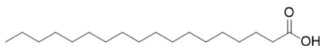
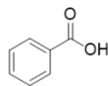
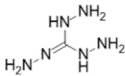
In thermogravimetric analysis, a brief increase in the recorded mass (or an unrealistically large mass loss) can occur even when the sample itself does not truly gain mass. This is mainly related to instrumental and gas-phase effects, including baseline drift and imperfect buoyancy correction as the purge-gas density changes with temperature, which can cause small systematic offsets in the microbalance signal. In addition, volatile products may re-condense on cooler parts of the pan/hanger or the furnace environment and then re-evaporate at higher temperatures, leading to distorted TG baselines and non-ideal apparent mass changes [20]. Therefore, such features should be treated as measurement artifacts unless they are accompanied by consistent chemical evidence (e.g., sustained uptake under reactive atmosphere) and reproducible behavior.

8.2.3 How can thermal analysis be cross validated with FTIR results, and what are common interpretation pitfalls?

FTIR provides functional-group fingerprints, while TG/DTG/DSC reveals thermal stability and phase/decomposition behavior; combining them improves identification reliability. For example, when FTIR indicates a long-chain saturated carboxylic acid structure, DSC is expected to show low-temperature endothermic transitions (melting and/or solid–solid rearrangements) that may occur with little or no mass change in TG, whereas the main TG mass loss should appear at higher temperatures due to volatilization and thermal decomposition under nitrogen [21]. A common pitfall is over-interpreting small DSC features as “decomposition” without corresponding TG mass loss or attributing minor TG anomalies to real chemistry without considering baseline/buoyancy artifacts and condensation–re-evaporation effects. Therefore, consistent interpretation should rely on the coupled evidence: (i) FTIR functional groups, (ii) whether TG shows a real mass-change step, and (iii) whether DSC endo/exotherms coincide with (or precede) DTG maxima in a physically reasonable manner [21].

9. Conclusion

This experimental group used infrared spectroscopy analysis and thermal analysis (TGA, DSC) methods to test and characterize the six samples and finally determined the specific types of the six samples. During the experiment, the group members deeply learned the basic principles of infrared spectroscopy and thermal analysis, and learned how to process data, draw graphs, and determine the corresponding functional groups based on the absorption peaks of the infrared spectrum, and how to judge the changes in the physical properties of the samples through the TG, DTG and DSC curves and use them to verify the results of the infrared spectroscopy. At the same time, the group members also learned why the infrared spectra obtained in the experiment differed from the standard spectra and the mathematical laws that different functional groups follow in their vibration frequencies. This experiment enabled us to better master the experimental operations of infrared spectroscopy and thermal analysis, laying a foundation for future sample structure characterization experiments. The final analysis results of the six samples are as follows.

Sample	Material	Structure
A1	Polyvinylpyrrolidone	
A2	Dopamine	
A3	Sodium dodecylbenzene sulfonate	
A4	Stearic acid	
A5	Benzoic acid	
A6	Tris (amino guanidinium) nitrate	

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